

RETRO-COMMISSIONING REPORT

IAH SPIRIT AIRLINES HANGAR

MARCH 20, 2023

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LIST OF ABBREVIATIONS

ABBREVIATION	TERM / PHRASE / NAME
AC	AIR-CONDITIONING UNIT
AHU	AIR-HANDLING UNIT
B	BOILER
BMcD	BURNS & McDONNELL
BMS	BUILDING MANAGEMENT SYSTEM
CH	CHILLER
CHWP	CHILLED WATER PUMP
CHWR	CHILLED WATER RETURN
CHWS	CHILLED WATER SUPPLY
CRAC	COMPUTER ROOM AIR-CONDITIONER
CU	CONDENSING UNIT
DDC	DIRECT DIGITAL CONTROLS
DX	DIRECT EXPANSION
EAT	ENTERING AIR TEMPERATURE
EF	EXHAUST FAN
EWT	ENTERING WATER TEMPERATURE
EXIST. / (E)	EXISTING
FCU	FAN COIL UNIT
FPB	FAN POWERED BOX
FPT	FUNCTIONAL TEST PROCEDURE
HAS	HOUSTON AIRPORT SYSTEM
HV	HEATING & VENTILATING UNIT
HVAC	HEATING, VENTILATION AND AIR CONDITIONING
HWP	HOT WATER PUMP
HWR	HOT WATER RETURN
HWS	HOT WATER SUPPLY
IAH	GEORGE BUSH INTERCONTINENTAL AIRPORT
LAT	LEAVING AIR TEMPERATURE
LWT	LEAVING WATER TEMPERATURE
MAT	MIXED AIR TEMPERATURE
MBH	1,000 BTU/HR
MC	MECHANICAL CONTRACTOR
MER	MECHANICAL EQUIPMENT ROOM
MFR	MANUFACTURER
O&M	OPERATIONS AND MAINTENANCE
OA	OUTDOOR AIR
RA	RETURN AIR
RCx	RETRO-COMMISSIONING
RTU	ROOFTOP UNIT
SA	SUPPLY AIR
SOO	SEQUENCE OF OPERATIONS
TAB	TESTING, ADJUSTING, AND BALANCING
UH	UNIT HEATER
VAV	VARIABLE AIR VOLUME
VFD	VARIABLE FREQUENCY DRIVE

Executive Summary

Spirit Airlines has retained the services of Burns & McDonnell (BMcD) to perform RCx services within Hangar and Warehouse buildings located at George Bush Intercontinental Airport in Houston, TX. The Retro-Commissioning process identifies problems and inefficiencies that have developed throughout a building's life due to aging equipment or changes to how occupants use spaces; and recommends solutions to ensure the facility runs at maximum efficiency. BMcD's RCx services included an on-site visual and functional inspection of existing HVAC equipment and a review of the Building Management System's (BMS) current programming.

This report aims to provide an overview of BMcD's findings, including deficiencies related to HVAC equipment and recommendations to utilize the building's BMS better, helping Spirit Airlines achieve optimal performance within the Hangar and Warehouse buildings.

The equipment recommendations are categorized as follows:

- Recommendation Types:
 - Design – Engineering Design documents are required to address issue.
 - Repair – Contractor to perform work including all permits to address issue.
- Priority Levels:
 - High – Required for the system to operate or to prevent future failure.
 - Medium – Required to improve system reliability.
 - Low – Required to improve system performance.

1.0 Hangar – RetroCx Findings

1.1 Chilled Water System: Chiller CH-1 & Pumps CHWP-1,2

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) CH-1	CHILLER	HANGAR ADMIN. AREA
HANGAR	(E) CHWP-1	PUMP	(E) CH-1
HANGAR	(E) CHWP-2	PUMP	(E) CH-1

The Hangar building's chilled water system consists of a Carrier pad-mounted outdoor air-cooled chiller (CH-1). According to its serial number, CH-1 is a 50-ton unit manufactured in 2013. The chilled water distribution system includes redundant constant speed pumps (CHWP-1 & CHWP-2) located in the first floor MER, and a single three-way control valve that serves the cooling coil for the office area AHU-1. Assuming regular maintenance, the system is expected to have another 10 years of useful service life.

The air-cooled chiller and associated chilled water pump(s) will enable during the following conditions: 1) when the outdoor air temperature is above the outdoor air temperature lockout setpoint (currently set to 45°F), and 2) when AHU-1's return air temperature is above AHU-1's return air temperature lockout setpoint (currently set to 68°F). Once the outdoor air and AHU-1 return air conditions are met, CHWP-1 shall enable. If CHWP-1 fails, CHWP-2 shall start. Once one of the chilled water pumps is proven "on", CH-1 shall enable and operate to supply a chilled water temperature of 45°F.

RCx Findings & Observations

- CH-1 – Air-Cooled Chiller
 - CH-1 was operational and properly cooling water to 45°F.
 - Changing of chilled water supply temperature is done via chiller's local controller.
 - BMS can start/stop CH-1.
 - No alarms or trending programmed in BMS. The only alarm from CH-1 to BMS is a loss of communication alarm. All other alarms are signaled outdoors at the chiller's local controller, including no/low flow & over/under voltage.
 - CH-1 cannot be manually reset at the local controller due to buttons on the controller being non-functional.
 - BMCD did not observe a previously known issue of cycling of the chiller's compressors.
 - Outdoor thermometer on the chilled water supply at CH-1 is not functional.
 - System is not provided with glycol or freeze protection controls.
- CHWP-1 & 2 – Chilled Water Pumps
 - CHWP-1 & 2 were operational and properly pumping water through the system.
 - When CH-1 is in "off" mode, the chilled water pumps can continue to run based on the current sequence of operations.
 - Currently, the chilled water pumps do not rotate between lead & stand-by.

1.2 Heating Hot Water System: Boiler B-1 & Pump HWP-1

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) B-1	BOILER	HANGAR ADMIN. AREA
HANGAR	(E) HWP-1	PUMP	(E) B-1

The Hangar building's gas-fired hot water heating boiler (B-1) and associated hot water pump (HWP-1) are located in Boiler Rm 136. The heating hot water system consists of a single 308 MBH hot water boiler, a single hot water distribution constant pump, and three-way control valves serving VAV heating coils throughout the office area. According to B-1's serial number, the manufacturing date is estimated to be 2006. Assuming regular maintenance, B-1 is expected to have another seven years of useful service life.

B-1 and associated HWP-1 will enable when the outdoor air temperature is below the outdoor air lockout setpoint (currently set to 40°F). Once the outdoor air conditions are met, HWP-1 shall enable. Once HWP-1 is proven "on", B-1 shall enable and operate to supply the required hot water temperature.

RCx Findings & Observations

- B-1 – Heating Hot Water Boiler
 - The existing B-1 does not have modulating capability, which will lead to cycling on & off. As a result, B-1 cannot provide a steady hot water supply temperature to the building's terminal VAV boxes and will cause fluctuations in space temperatures. B-1's lack of modulating capability will be most apparent during the Spring & Fall seasons when full heating is not required.
 - Currently, the hot water supply temperature is set to 160°F. Per the original design, the HWS temperature should be set to 180°F. The HWS temperature is changed via an integral mechanical knob in B-1's access panel.
 - BMS does not monitor B-1 status.
 - BMS monitors HWS/R temperatures.
 - No alarms or trending programmed to BMS. The only alarm from chiller to BMS is a loss of communication alarm. All other alarms are signaled outdoors at the chiller's local controller, including no/low flow & over/under voltage.
 - Per the current & original VAV sequence of operations, the VAV boxes' primary inlet damper will close when the zone calls for heating. This results in a reduced building heating load required by the boiler as well as unventilated spaces during the winter season. Refer to section 1.4, Admin. Area: VAVs, for more testing findings & observations of Administration Area's terminal VAV boxes.
 - Per the current AHU-1 sequence of operations, the AHU's outdoor air damper will close when the outdoor air temperature is below 50°F. This reduces the building heating load the boiler and unventilated spaces require during the winter season. Refer to section 1.3, Admin. Area: AHU-1, for more testing findings & observations of AHU-1.
 - The existing B-1 has internal safeties for boiler shutdown.
- HWP-1 – Hot Water Pump
 - HWP-1 was operational, properly pumping water through the system.
 - Currently, only one hot water pump is installed in the system and does not offer redundancy in the case of pump failure.

1.3 Admin Area: AHU-1

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) AHU-1	AIR-HANDLING UNIT	HANGAR ADMIN. AREA

The existing air handling unit (AHU-1) in Mechanical Room 130 serves the Administration Area of the Hangar building. The estimated installation year of AHU-1 is 2013, and assuming proper maintenance, is within its useful life of 25 years. AHU-1 operates as a variable-air-volume system and utilizes terminal VAV boxes to modulate airflow to zones within the Administration Area.

When AHU-1 is enabled, the outdoor air damper will open to maintain the minimum required outdoor airflow rate. However, currently, the BMS commands the OA damper to close when the OA temperature is below 50°F. AHU-1's variable frequency drive (VFD) will modulate the supply fan to maintain a static pressure setpoint via a static pressure sensor in the supply air main duct. AHU-1's chilled water valve will modulate to maintain the supply air temperature (SAT) setpoint. Currently, AHU-1 utilizes a supply air temperature sequence that will reset the supply air temperature (with a dead band) based on the following:

- When AHU-1's return air temperature is above 70°F, the unit's SAT shall be set to 53°F.
- When AHU-1's return air temperature is below 70°F, the unit's SAT shall be set to 60°F.

RCx Findings & Observations

- AHU-1 – Air-Handling Unit
 - As currently programmed, BMS closes the OA damper when the OA temperature is below 50°F. As a result, spaces served by AHU-1 will not be ventilated when associated VAV boxes go into heating mode.
 - OA damper associated with AHU-1 is stuck in an open position, possibly from a malfunctioning damper actuator.
 - BMS does not monitor supply fan VFD speed.
 - AHU-1's static pressure sensor controlling supply fan is in Mechanical Room 130 in the supply main serving the first floor.
 - AHU-1 does not have a preheat coil.
 - The CHW valve modulates correctly to maintain the supply air temperature. BMS does not monitor CHW valve status.
 - AHU-1 does not utilize an economizer mode.
 - AHU-1 does not monitor space relative humidity.
 - No alarms or trending programmed in BMS.
 - AHU-1 currently does not run based on the building's occupancy schedule.
 - AHU-1's filter is not monitored for "dirty" status.
 - Four duct smoke detectors (2 in SA mains, 2 in RA mains) are located in AHU-1's ductwork.
 - Freezestat is not installed in AHU-1. Installation of a freezestat will shut down AHU-1 based on low entering air temperature to the coil.
 - The BMS OA sensor is not reading correctly. The sensor reads approximately 15°F above the actual OA ambient temperature.

1.4 Admin Area: VAV Boxes, VAV-1 to 23

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) VAV-01	FAN-POWERED BOX	LOBBY 108
HANGAR	(E) VAV-02	FAN-POWERED BOX	SECURITY 109
HANGAR	(E) VAV-03	FAN-POWERED BOX	OFFICE 112
HANGAR	(E) VAV-04	FAN-POWERED BOX	KITCHEN 118
HANGAR	(E) VAV-05	FAN-POWERED BOX	ADMIN GROUND FLR CORRIDOR
HANGAR	(E) VAV-06	FAN-POWERED BOX	SHOP 128
HANGAR	(E) VAV-07	FAN-POWERED BOX	SHOP 128
HANGAR	(E) VAV-08	FAN-POWERED BOX	SHOP 128
HANGAR	(E) VAV-09	FAN-POWERED BOX	ADMIN GROUND FLR CORRIDOR
HANGAR	(E) VAV-10	FAN-POWERED BOX	OFFICE 202
HANGAR	(E) VAV-11	FAN-POWERED BOX	OFFICE 202
HANGAR	(E) VAV-12	FAN-POWERED BOX	OFFICE 202
HANGAR	(E) VAV-13	FAN-POWERED BOX	OFFICE 202
HANGAR	(E) VAV-14	FAN-POWERED BOX	OFFICE 208
HANGAR	(E) VAV-15	FAN-POWERED BOX	FLIGHT OPERATIONS 217
HANGAR	(E) VAV-16	FAN-POWERED BOX	ACCOUNTING 216
HANGAR	(E) VAV-17	FAN-POWERED BOX	COFFEE 219
HANGAR	(E) VAV-18	FAN-POWERED BOX	CONF RM 218
HANGAR	(E) VAV-19	FAN-POWERED BOX	TRAINING DEVELOPMENT 224
HANGAR	(E) VAV-20	FAN-POWERED BOX	PILOTS OFFICE
HANGAR	(E) VAV-21	FAN-POWERED BOX	OFFICE 230
HANGAR	(E) VAV-22	FAN-POWERED BOX	IT RM 231
HANGAR	(E) VAV-23	VAV BOX	SHOP 128

23 Variable Air Volume (VAV) terminal units serve the Administration Area of the Hangar Building. VAVs 1 through 22 are fan-powered VAV boxes with hot water reheat coils. The fans in these boxes are arranged in a “parallel” configuration. VAV-23 is a cooling-only terminal unit that serves Hangar Restroom Room 100B.

All zones have a BMS space temperature setpoint of 72°F. Local zone thermostats associated with each VAV box have adjustable dials to change the space temperature setpoint +/- 3°F from the BMS setpoint. The current VAV Box Sequence of Operations is as follows:

- VAV-1 thru 22 (Parallel Fan-Powered VAV):
 - When in cooling mode, the integral fan shall disable, and the motorized damper at the box’s inlet shall modulate between 0%-100% to maintain the space temperature setpoint.
 - When in heating mode, the integral fan shall energize and supply full airflow at 100% capacity. The motorized damper at the box’s inlet shall close. The hot water valve shall modulate to maintain the space temperature setpoint.
- VAV-23 (Cooling only VAV):
 - The VAV’s motorized damper at the box’s inlet shall modulate between 0%-100% to maintain the space temperature setpoint.

RCx Findings & Observations

- VAVs – General Findings
 - VAV Boxes are not balanced to maintain a minimum or maximum airflow setpoint. VAV damper will modulate between 0%-100%.

- During heating mode, VAVs will close the inlet damper. As a result, spaces receive no ventilation air when the VAV box is in heating mode.
 - Currently programmed is a single space setpoint of 72°F for all spaces. BMS does not change space temperature setpoint based on building occupancy schedule or season.
 - VAVs 1 thru 22 do not have airflow measuring capability. VAV-23 has an airflow measuring device but does not read airflow in BMS.
 - HW reheat valves do not provide BMS valve position feedback.
 - Discharge air temperature sensors are not installed on VAVs 1 thru 22.
 - No alarms or trending programmed in BMS
- VAV-1 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to “VAVs – General Findings” for further information.
 - VAV-2 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to “VAVs – General Findings” for further information.
 - VAV-3 – Fan-Powered VAV Box w/ HW Reheat Coil
 - VAV box was originally named VAV-30 in BMS. Controls updated during testing to rename this box to VAV-3
 - Refer to “VAVs – General Findings” for additional information.
 - VAV-4 – Fan-Powered VAV Box w/ HW Reheat Coil
 - During testing in cooling mode, the supply air temperature recorded was approximately 68°F. At this time, AHU-1 was supplying air at ~53°F.
 - Refer to “VAVs – General Findings” for additional information.
 - VAV-5 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to “VAVs – General Findings” for further information.
 - VAV-6 – Fan-Powered VAV Box w/ HW Reheat Coil
 - During testing in heating mode, the VAV fan did not energize. As a result, heating functionality could not be confirmed for VAV-6.
 - The valve actuator appeared to be functional.
 - Refer to “VAVs – General Findings” for additional information.
 - VAV-7 – Fan-Powered VAV Box w/ HW Reheat Coil
 - During testing in heating mode, the VAV fan did not energize. As a result, heating functionality could not be confirmed for VAV-7.
 - The valve actuator did not appear to be functional.
 - Refer to “VAVs – General Findings” for additional information.
 - VAV-8 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to “VAVs – General Findings” for further information.

- VAV-9 – Fan-Powered VAV Box w/ HW Reheat Coil
 - During testing in heating mode, VAV-9's fan energized, and the valve actuator appeared to be functional. However, the discharge air temperature did not rise above 72°F.
 - Refer to "VAVs – General Findings" for additional information.
- VAV-10 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to "VAVs – General Findings" for further information.
- VAV-11 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to "VAVs – General Findings" for further information.
- VAV-12 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to "VAVs – General Findings" for further information.
- VAV-13 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to "VAVs – General Findings" for further information.
- VAV-14 – Fan-Powered VAV Box w/ HW Reheat Coil
 - VAV box was originally named VAV-3 in BMS. Controls updated during testing to rename this box to VAV-14
 - Could not verify heating functionality of VAV-14 as the hot water heating boiler was off during testing.
 - Refer to "VAVs – General Findings" for additional information.
- VAV-15 – Fan-Powered VAV Box w/ HW Reheat Coil
 - During testing in cooling mode, the supply air temperature recorded was approximately 68°F. At this time, AHU-1 was supplying air at ≈53°F.
 - Refer to "VAVs – General Findings" for additional information.
- VAV-16 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to "VAVs – General Findings" for further information.
- VAV-17 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to "VAVs – General Findings" for further information.
- VAV-18 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to "VAVs – General Findings" for further information.
- VAV-19 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to "VAVs – General Findings" for further information.

- VAV-20 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to “VAVs – General Findings” for further information.
- VAV-21 – Fan-Powered VAV Box w/ HW Reheat Coil
 - No additional Findings. Refer to “VAVs – General Findings” for further information.
- VAV-22 – Fan-Powered VAV Box w/ HW Reheat Coil
 - VAV-22 will be replaced as part of the Hangar Repair Project. Notes are provided for reference only.
 - During testing in heating mode, VAV-22’s fan energized, and the valve actuator appeared to be functional. However, the discharge air temperature did not rise above 72°F.
 - Refer to “VAVs – General Findings” for additional information.
- VAV-23 – VAV Box (Cooling Only)
 - VAV box is currently named “Hangar Restroom” in BMS. The box should be renamed to VAV-23.
 - VAV-23 has an airflow measuring device at the inlet of the box. However, BMS is currently not monitoring airflow.
 - Refer to “VAVs – General Findings” for additional information.

1.5 Server Room: CRAC-1 (CU-6), CRAC-2 (CU-5) & AHU-2

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) CRAC-1	COMPUTER ROOM UNIT	IT RM 231
HANGAR	(E) CRAC-2	COMPUTER ROOM UNIT	IT RM 231
HANGAR	(E) AHU-2	AIR-HANDLING UNIT	IT RM 231
HANGAR	(E) CU-5	CONDENSING UNIT	(E) CRAC-2
HANGAR	(E) CU-6	CONDENSING UNIT	(E) CRAC-1

Three systems currently serve Server Room 231. These systems include:

- Two floor mounted Data Aire units, CRAC-1 (CU-6) & CRAC-2 (CU-5)
- Packaged Trane Rooftop DX unit, AHU-2

CRACs 1 & 2, located in Server Room 231, are each 10-ton units with two cooling stages. Their associated condensing units are on the roof. CRACs 1 & 2 are approximately 13 years old and nearing their 15-year service life. Each CRAC has a humidifier but is not connected to a water line. They each have an internal condensate pump and an integral thermostat & humidity sensor adjustable via their on-board displays. CRACs 1 & 2 will cycle to maintain space temperature and humidity setpoints.

The Trane cooling-only packaged rooftop unit (RTU), tagged AHU-2 and manufactured in 2017, is 10 tons. AHU-2 cycles to maintain the space temperature setpoint and is controlled via a local wall-mounted thermostat in Server Room 231.

As noted during the condition assessment, the total cooling capacity of the equipment (over 30 tons) is significantly larger than the required cooling capacity (5 tons). System oversizing

will cause the system to cycle continually, decreasing the useful life of the equipment. The VAV-22 system is to be demolished to reduce the operating cooling capacity, and only a single unit is to operate at a time. Currently, AHU-2 is the primary cooling equipment for the room. CRACs 1 & 2 were not operable during testing, as indicated below.

RCx Findings & Observations

- CRAC-1 (CU-6) – Computer Room Air-Conditioner
 - The on-board display panel was not functional.
 - When calling for cooling, unit compressors did not enable.
 - i. Could not test the cooling functionality of the system.
 - ii. Could not test the dehumidification mode of the system.
 - iii. Could not test the functionality of condensing unit CU-6.
 - iv. Could not test the functionality of the internal condensate pump.
 - BMS does not monitor the unit.
 - CRAC-1 appears to be oversized with the current equipment load in the space.
- CRAC-2 (CU-5) – Computer Room Air-Conditioner
 - When calling for cooling, unit compressors did not enable.
 - i. Could not test the cooling functionality of the system.
 - ii. Could not test the dehumidification mode of the system.
 - iii. Could not test the functionality of condensing unit CU-5.
 - iv. Could not test the functionality of the internal condensate pump.
 - BMS does not monitor the unit.
 - CRAC-2 appears to be oversized with the current equipment load in the space.
- AHU-2 – Rooftop Unit
 - BMS does not monitor the unit.
 - AHU-2 appears to be oversized with the current equipment load in the space. AHU-2 cooled the space from 72°F to 65°F in under 10 minutes.

1.6 Hangar: GRH'S (20 TOTAL)

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) GRH	GAS RADIANT HEATER	HANGAR

There are 20 existing gas-fired radiant heaters (GRHs) in the Hangar, controlled by four separate zones. Refer to Appendix C for the GRH Zoning Plan. A local wall-mounted thermostat controls each zone within the Hangar. When the space temperature drops below the thermostat setting, all GRHs associated with that zone are enabled.

RCx Findings & Observations

- GRH Zone 1 (6 GRHs) – South-East Zone
 - All GRHs in Zone 1 are operational. No issues found.
 - The local wall-mounted thermostat is functional but in poor condition.



- GRH Zone 2 (6 GRHs) – North-East Zone
 - Five of six GRHs in Zone 2 did not enable when commanded.
 - The local wall-mounted thermostat is functional but in poor condition and missing parts.
- GRH Zone 3 (4 GRHs) – North-West Zone
 - One of four GRHs in Zone 3 did not enable when commanded.
 - The local wall-mounted thermostat is functional but in poor condition and missing parts.
- GRH Zone 4 (4 GRHs) – South-West Zone
 - One of four GRHs in Zone 4 did not enable when commanded.
 - The local wall-mounted thermostat is functional but in poor condition.

1.7 2nd Fl. Storage Rm 236: HV-6 (CU-2) & EF-10

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) HV-6	HEATING VENTILATION UNIT	STORAGE 235
HANGAR	(E) CU-2	CONDENSING UNIT	HV-6
HANGAR	(E) EF-10	EXHAUST FAN	STORAGE 235

The Heating Ventilation Unit (HV-6) in Storage Room 236 includes an R-22 DX cooling coil and a hot water heating coil. The associated outdoor condensing unit (CU-2) is on the north side of the Hangar building. HV-6 recirculates & conditions air and does not provide ventilation to the space. Condensate is drained via gravity outdoors. HV-6 is controlled via a local wall-mounted thermostat and cycles to maintain the heating & cooling space temperature setpoints.

EF-10 is a wall-mounted, propellor-type exhaust fan on the north exterior wall of Storage Room 236. When operating, outdoor air is brought into the space via a backdraft damper at the OA louver and exhausted out via EF-10. A local wall-mounted thermostat controls EF-10. When the temperature rises above the setpoint, EF-10 enables.

Note: This unit originally operated as a heating & ventilation unit only. The unit currently operates as an air-conditioning unit and no longer provides ventilation to the space. This unit is labeled "SECOND FLOOR AC" on the BMS graphics.

RCx Findings & Observations

- HV-6 (CU-2)
 - The unit is functional and cooling the space properly.
 - Unit HW control valve is functional. Could not test HV-6 heating mode as HW heating boiler was not operational during testing.
 - Currently, the space is not ventilated; the unit is recirculating only.
 - Condensate from the unit is drained outdoors via gravity. Not observed was a condensate overflow switch. A condensate overflow should shut down the unit in the event the main drain line is blocked to prevent water damage to the building and equipment in the space.
 - BMS currently monitors the space temperature of Storage Room 236.
 - No alarms or trending programmed in BMS.
 - Currently, the unit maintains a single setpoint for both heating & cooling (72°F).

- EF-10 – Wall-mounted exhaust fan.
 - EF-10 was not operational during testing.
 - It is assumed exhaust fan EF-10 was designed to ventilate the space when the space temperature rose above the desired space temperature during the summer months. Since HV-6 includes mechanical cooling, EF-10 should not be placed into operation.

1.8 Pump Rm 100B: UH-1 & EF-13

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) UH-1	UNIT HEATER	PUMP RM 100B
HANGAR	(E) EF-13	EXHAUST FAN	PUMP RM 100B

The existing gas-fired unit heater (UH-1) serves Pump Room 100B. UH-1 will cycle on/off via a local wall-mounted thermostat in Pump Room 100B to maintain the space temperature setpoint.

Exhaust fan (EF-13) provides natural cooling ventilation to Pump Room 100B. The system includes a motorized control damper at a wall-mounted OA louver for intake. EF-13 will cycle on/off via a local wall-mounted thermostat in Pump Room 100B. The motorized OA control damper will open when the space temperature rises above the setpoint. Once the control damper opens, EF-13 will enable.

RCx Findings & Observations

- UH-1 – Unit Heater
 - UH-1 is operational. No issues found.
 - BMS does not monitor the unit/space.
- EF-13 – Exhaust Fan
 - EF-13 is operational. No issues found.
 - The motorized control damper was not functional and was stuck in the open position. The actuator attempts to close, but the damper blades remain open.

1.9 Hangar Restroom Rm 100A: UH-2 & EF-14

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) UH-2	UNIT HEATER	HANGAR RESTROOM 100A
HANGAR	(E) EF-14	EXHAUST FAN	HANGAR RESTROOM 100A

The existing wall-mounted electric unit heater (UH-2), serving Hangar Restroom 100A, will cycle on/off via an integral thermostat to maintain the space temperature setpoint.

Exhaust fan (EF-14) exhausts Hangar Restroom 100A and is controlled via a local wall-mounted occupancy sensor.

RCx Findings & Observations

- UH-2 – Unit Heater
 - UH-2 is not operational. The electric resistance heating element enabled, but the fan did not energize.
 - BMS does not monitor the unit/space.
- EF-14 – Exhaust Fan
 - EF-14 is operational. No issues found.
 - EF-14 was missing ceiling grille during testing.

1.10 EF-8

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) EF-08	EXHAUST FAN	ELEC. RM 233

Exhaust fan (EF-8) exhausts Electrical Room 233 and runs continuously 24/7/365.

RCx Findings & Observations

- EF-8 – Exhaust Fan
 - EF-8 is operational. No issues found.

1.11 EF-9

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) EF-09	EXHAUST FAN	RM 133 & RM 132

Exhaust fan (EF-9) exhausts Rooms 132 and 133 on the first floor and runs continuously 24/7/365.

RCx Findings & Observations

- EF-9 – Exhaust Fan
 - EF-9 is operational. No issues found.



2.0 Warehouse – RetroCx Findings

2.1 Warehouse: UH-1 thru 7

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
WAREHOUSE	(E) UH-1-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-2-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-3-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-4-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-5-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-6-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-7-W	UNIT HEATER	100 WAREHOUSE

The existing gas-fired unit heaters in the Warehouse, supported by the structure, provide heat to the space. The unit heaters will cycle on/off via local wall-mounted thermostats in the Warehouse to maintain the space temperature setpoint.

RCx Findings & Observations

- UH-1-W
 - UH-1-W is operational. No issues found.
- UH-2-W
 - UH-2-W is operational. No issues found.
- UH-3-W
 - UH-3-W is operational. No issues found.
- UH-4-W
 - UH-4-W is operational. No issues found.
- UH-5-W
 - UH-5-W is operational. No issues found.
- UH-6-W
 - When calling for heating, the UH-6-W fan enabled, but the burner did not fire.
- UH-7-W
 - UH-7-W is operational. No issues found.

2.2 Admin Comm. Rm 111: AC-1-W (CU-1-W)

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
WAREHOUSE	(E) AC-1-W	MINI-SPLIT AC	111 COMM.
WAREHOUSE	(E) CU-1-W	CONDENSING UNIT	(E)AC-1-2

The existing wall-mounted mini-split air-conditioning unit (AC-1-W) serves Communications Room 111 on the 1st floor. The associated condensing unit (CU-1-W) is on the roof of the 1st floor Women's restroom. AC-1-W is controlled by a local thermostat/controller in Room 111 and cycles to maintain the space temperature setpoint.

RCx Findings & Observations

- AC-1-W & CU-1-W
 - The system is operational and properly cools the space.
 - Indoor unit airflow deflectors had trouble opening & closing. Deflectors got stuck in the closed position when enabling the unit from off mode. Deflectors had to be manually opened.
 - Condensate from the indoor unit is drained via gravity to a floor drain in the adjacent room. The unit does not utilize a condensate pump, and a condensate overflow switch was not observed. A condensate overflow should shut down the unit in the event the main drain line is blocked to prevent water damage to the building and equipment within the space.
 - BMS does not currently monitor the existing split-system.
 - During RCx, a sign was observed on the entrance to Comm. Room 111 stating, "Please Leave Door Open". This door must remain closed to ensure the split-system AC maintains the space temperature setpoint required by the comm. equipment.

2.3 Admin 1st Fl. Restrooms 103 & 104: EF-1-W

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
WAREHOUSE	(E) EF-1-W	EXHAUST FAN	1ST FL. RESTROOMS

The existing inline exhaust fan (EF-1-W) serves the Men's & Women's restrooms on the first floor. The exhaust fan is interlocked to AHU-1-W and runs continuously when the air-handling unit is enabled.

RCx Findings & Observations

- EF-1-W
 - EF-1-W is operational. No issues found.

2.4 Admin 2nd Fl. Restroom 208: EF-2-W

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
WAREHOUSE	(E) EF-2-W	EXHAUST FAN	2ND FL. RESTROOM

A wall-mounted occupancy sensor controls the ceiling exhaust fan (EF-2-W) in the 2nd-floor single-occupancy restroom.

RCx Findings & Observations

- EF-2-W
 - EF-2-W is operational. No issues found



3.0 Hangar Recommendations

HANGAR BLDG. - CHILLED WATER SYSTEM				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) CH-1	CHILLER	REPAIR	MEDIUM	ADD ALARMS TO BE MONITORED BY BMS. AT A MINIMUM, ALARMING BASED ON HIGH CHILLED WATER SUPPLY TEMPERATURE TO ALERT IF CHILLER IS DOWN.
		REPAIR	HIGH	REPAIR LOCAL CONTROLLER TO ALLOW LOCAL MANUAL RESET.
		REPAIR	LOW	REPLACE THERMOMETER.
		DESIGN	MEDIUM	PROVIDE FREEZE PROTECTION CONTROLS WHEN AMBIENT TEMPERATURE IS BELOW 35°F.
(E) CHWP-1 & (E) CHWP-2	PUMP	DESIGN	LOW	UPDATE BMS PROGRAMMING TO OPTIMIZE THE CHILLED WATER SYSTEMS SEQUENCE OF OPERATIONS.

HANGAR BLDG. - HOT WATER SYSTEM				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) B-1	BOILER	DESIGN	HIGH	REPLACE EXISTING HEATING HOT WATER BOILER WITH A BOILER HAVING MODULATING CAPABILITY SIZED FOR BUILDING'S CALCULATED HEATING LOADS. NEW BOILER SHALL BE DESIGNED WITH OPTIMIZED CONTROLS, INCLUDING ALARMS & SAFETIES, AND INTEGRATED TO THE BMS.
(E) HWP-1	PUMP	DESIGN	MEDIUM	INSTALL A SECOND HOT WATER PUMP FOR SYSTEM REDUNDANCY.

HANGAR BLDG. - GAS-RADIANT HEATERS				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) GRH	GAS RADIANT HEATER	REPAIR	HIGH	INVESTIGATE RADIANT HEATERS THAT DID NOT ENABLE WHEN COMMANDED. REPAIR / REPLACE AS REQUIRED. REFER TO RADIANT HEATER ZONE PLAN IN APPENDIX C FOR ADDITIONAL INFORMATION.
		REPAIR	MEDIUM	REPLACE WALL MOUNTED THERMOSTATS.

HANGAR BLDG. - HVAC SYSTEM (AHU-1)				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) AHU-1	AIR-HANDLING UNIT	REPAIR	HIGH	UPDATE CONTROLS TO PROVIDE MINIMUM VENTILATION AIRFLOW (2,500 CFM) WHEN BUILDING IS OCCUPIED. TESTING AND BALANCING SPECIALIST TO DETERMINE OA DAMPER OPEN POSITION SETTING.
		DESIGN/ REPAIR	HIGH	REPAIR - REPLACE/REPAIR OA CONTROL DAMPER AS REQUIRED DESIGN (OPTION) - PROVIDE AIR FLOW MEASURING STATION TO ENSURE PROPER VENTILATION AIRFLOW IS PROVIDED.
		REPAIR	HIGH	PROVIDE SUN/WEATHER SHIELD AS REQUIRED FOR ACCURATE CONTROL BASED ON ACCURATE AMBIENT TEMPERATURES.
		DESIGN	HIGH	PROVIDE FREEZESTAT TO PROTECT CHILLED WATER CONTROL VALVE. FREEZESTAT WILL SHUT DOWN AHU-1 AND SIGNAL ALARM TO BMS.
		REPAIR	MEDIUM	PROVIDE "DIRTY FILTER" ALARM BASED ON SUPPLY FAN RUN TIME HOURS.
		DESIGN	MEDIUM	UPDATE AHU'S SEQUENCE TO INCLUDE OCCUPANCY SCHEDULE AND TRENDING INFORMATION TO ENHANCE OCCUPANCY COMFORT AND ENERGY EFFICIENCY.

HANGAR BLDG. - VAV'S				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) VAVs	ALL VAVs	DESIGN/ REPAIR	HIGH	REPAIR - TESTING & BALANCING SPECIALIST TO DETERMINE MIN & MAX DAMPER POSITIONS TO PROVIDE MIN & MAX AIRFLOW SETPOINTS. DESIGN (OPTION) - INSTALL AIRFLOW MEASURING DEVICE AT EACH VAV BOX INLET FOR CONTROL OF BOXES TO THEIR MIN & MAX AIRFLOW SETPOINTS
		REPAIR	HIGH	UPDATE PROGRAMMING FOR VAV BOX DAMPER TO GO TO MINIMUM POSITION DURING HEATING MODE AS DETERMINED BY TESTING & BALANCING SPECIALIST.
		DESIGN	MEDIUM	UPDATE VAV BOX SEQUENCE OF OPERATIONS TO INCLUDE OCCUPANCY SCHEDULE AND TRENDING INFORMATION TO ENHANCE OCCUPANCY COMFORT AND ENERGY EFFICIENCY.
(E) VAV-04	FAN-POWERED BOX	REPAIR	HIGH	TESTING & BALANCING SPECIALIST TO INVESTIGATE THE AIR SUPPLY AIRFLOW FROM VAV-04. VERIFY DAMPER IS FULLY OPERATIONAL.
(E) VAV-06	FAN-POWERED BOX	REPAIR	HIGH	REPLACE FAN MOTOR / FAN POWERED BOX AS REQUIRED.
(E) VAV-07	FAN-POWERED BOX	REPAIR	HIGH	REPLACE FAN MOTOR / FAN POWERED BOX & HOT WATER CONTROL VALVE AS REQUIRED.
(E) VAV-09	FAN-POWERED BOX	REPAIR	HIGH	INVESTIGATE ISOLATION VALVE, HW PIPING, AND STRAINER TO VERIFY FLOW IS NOT OBSTRUCTED TO FIX HEATING ISSUE.
(E) VAV-14	FAN-POWERED BOX	REPAIR	MEDIUM	SET-UP TREND INFORMATION FOR VAV-14 TO MONITOR. VERIFY SYSTEM PERFORMANCE.
(E) VAV-15	FAN-POWERED BOX	REPAIR	HIGH	TESTING & BALANCING SPECIALIST TO INVESTIGATE THE AIR SUPPLY AIRFLOW FROM VAV-15. VERIFY DAMPER FULLY OPERATIONAL.
(E) VAV-23	VAV BOX	REPAIR	MEDIUM	RENAME BOX TO VAV-23.
		REPAIR	LOW	UPDATE VAV PROGRAMMING TO INCLUDE MONITORING OF VAV AIRFLOW MEASURING STATION. CALIBRATE AIRFLOW FOR PROPER OPERATION.

HANGAR BLDG. - SERVER ROOM 231				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) CRAC-1 & (E) CU-6	COMPUTER ROOM UNIT & CONDENSING UNIT	REPAIR	MEDIUM	FURTHER INVESTIGATE MALFUNCTIONING CRAC COMPRESSOR. REPAIR / REPLACE AS NECESSARY.
		REPAIR	MEDIUM	PROVIDE SPACE TEMPERATURE SENSOR FOR SERVER ROOM FOR MONITORING BY THE BMS.
		REPAIR	MEDIUM	PROVIDE MANUFACTURER REPRESENTATIVE TO INVESTIGATE AND REPAIR AS REQUIRED.
(E) CRAC-2 & (E) CU-5	COMPUTER ROOM UNIT & CONDENSING UNIT	REPAIR	MEDIUM	PROVIDE MANUFACTURER REPRESENTATIVE TO INVESTIGATE AND REPAIR AS REQUIRED.
(E) AHU-2	AIR-HANDLING UNIT	REPAIR	LOW	BUILDING FACILITY TO MONITOR AHU-2 AND SPACE TO MAKE SURE UNIT DOES NOT CONTINUOUSLY CYCLE.

HANGAR BLDG. - HV-6 ROOM 235				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) HV-6 & (E) CU-2	HEATING VENTILATION UNIT & CONDENSING UNIT	REPAIR	LOW	SET UP TRENDING OF SPACE TEMPERATURE SENSOR TO VERIFY PROPER OPERATION OF HOT WATER CONTROL VALVE.
		REPAIR	LOW	PROGRAM ALARM WHEN SPACE TEMPERATURE DROPS BELOW OR RISES ABOVE SETPOINT.
		REPAIR	LOW	PROGRAM SEPARATE HEATING AND COOLING SETPOINTS FOR ENERGY CONSERVATION.
		REPAIR	LOW	VERIFY INSTALLATION OF CONDENSATE OVERFLOW SWITCH. PROVIDE IF NOT INSTALLED.
(E) EF-10	EXHAUST FAN	REPAIR	LOW	DECOMMISSION AND/OR DEMOLISH EF-10.

HANGAR BLDG. - PUMP ROOM 100B				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) UH-1	UNIT HEATER	REPAIR	HIGH	INSTALL SPACE TEMPERATURE SENSOR MONITORED BY BMS. PROGRAM BMS TO ALARM BASED ON HIGH AND LOW SPACE TEMPERATURE.
(E) EF-13	EXHAUST FAN	REPAIR	HIGH	REPAIR / REPLACE MOTORIZED CONTROL DAMPER

HANGAR BLDG. - HANGAR RESTROOM 100A				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) UH-2	UNIT HEATER	REPAIR	HIGH	REPLACE ELECTRIC UNIT HEATER.
		REPAIR	HIGH	PROGRAM BMS TO ALARM BASED ON LOW & HIGH SPACE TEMPERATURE. UTILIZE SPACE TEMPERATURE SENSOR ON VAV-23.
(E) EF-14	EXHAUST FAN	REPAIR	LOW	INSTALL / REPLACE EXHAUST FAN CEILING GRILLE.



4.0 Warehouse Recommendations

WAREHOUSE BLDG. - COMM. ROOM 111				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) AC-1-W & (E) CU-1-W	MINI-SPLIT AC UNIT & CONDITIONING UNIT	REPAIR	MEDIUM	INVESTIGATE TO ENSURE DEFLECTORS OPEN WHEN CALLED FOR COOLING. REPAIR AS REQUIRED.
		REPAIR	MEDIUM	VERIFY INSTALLATION OF CONDENSATE OVERFLOW SWITCH. INSTALL NEW OVERFLOW SWITCH IF NECESSARY.
		REPAIR	MEDIUM	INSTALL OF SPACE TEMPERATURE SENSOR MONITORED BY BMS. PROGRAM BMS TO ALARM BASED ON HIGH SPACE TEMPERATURE.
		REPAIR	MEDIUM	REMOVE SIGN AND KEEP DOOR CLOSED FOR PROPER OPERATION.

WAREHOUSE BLDG. - WAREHOUSE				
UNIT	EQUIPMENT TYPE	RECOMM. TYPE	PRIORITY	RECOMMENDATION DESCRIPTION
(E) UH-6-W	UNIT HEATER	REPAIR	MEDIUM	INVESTIGATE HEATER FUNCTIONALITY. REPLACE UNIT OR REPAIR BURNER ASSEMBLY AS REQUIRED.

APPENDIX A HANGAR EQUIPMENT PLANS

FILE PATH: Autodesk Docs://147053_SPIRITIAHHANG/147053-Mech-Hangar.rvt

HAS FILE:

PLOT DATE:
DOA DWG FILE:
OLD DOA No.:

1 MECHANICAL HANGAR OVERALL PLAN - FIRST FLOOR

1/16" = 1'-0"



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16875 JOHN F. KENNEDY BLVD.
HOUSTON, TX 77032

SPIRIT AIRLINES -
IAH HANGAR RENOVATION

T.I.P. No.	TIP-22-177-IAH	A.I.P. No.	N/A
C.O.H. No.	N/A	D.O.A No.	N/A

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LICENSE NO. F-A45

DESIGNER PROJECT No.: 147053

PROJECT STATUS: CD

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No.	DESCRIPTION	DATE	BY
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DESIGNER:	Designer
DRAWN BY:	Author
CHECKED BY:	Checker
ISSUE DATE:	03/08/23
APPROVED BY:	Approver
APPROVAL DATE:	-

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of
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SHEET NAME: MECHANICAL 1ST FL PLAN

SHEET No. 23-EQP-101 SCALE: 1/16" = 1'-0"

SHEET SIZE: 22"x34" ANSI-D

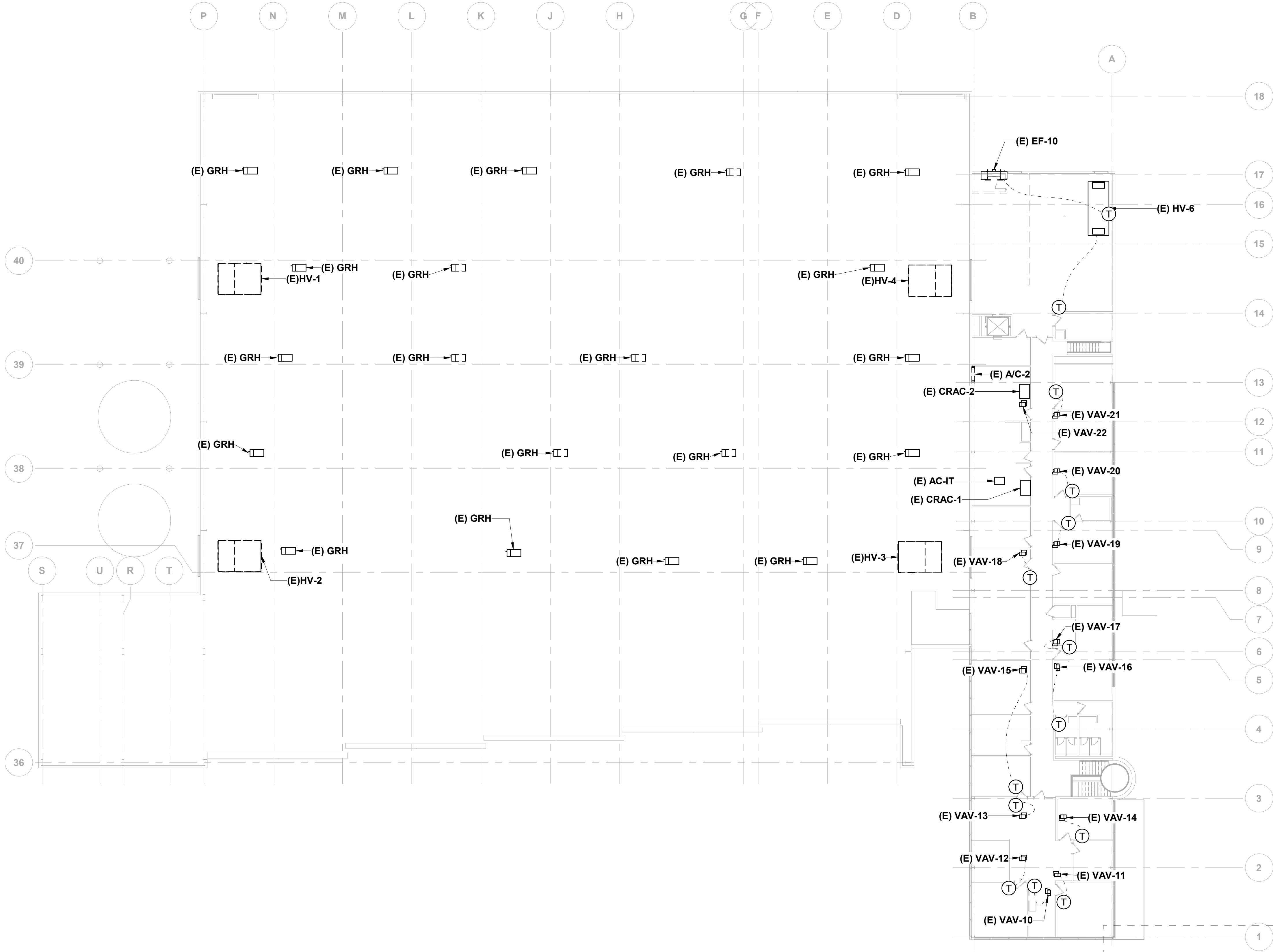
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FILE PATH: Autodesk Docs://147053_SPIRITIAHHANG/147053-Mech-Hangar.rvt

HAS FILE:

PLOT DATE:
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1 MECHANICAL HANGAR OVERALL PLAN - SECOND FLOOR



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SHEET NAME: MECHANICAL 2ND FL PLAN

SHEET No. 23-EQP-102 SCALE: 1/16" = 1'-0"

SHEET SIZE: 22"x34" ANSI-D

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FILE PATH: Autodesk Docs://147053_SPIRITIAHHANG/147053-Mech-Hangar.rvt

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1 MECHANICAL HANGAR OVERALL PLAN - ROOF

1/16" = 1'-0"



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SHEET NAME: MECHANICAL ROOF PLAN

SHEET No. 23-EQP-103 SCALE: 1/16" = 1'-0"

SHEET SIZE: 22"x34" ANSI-D

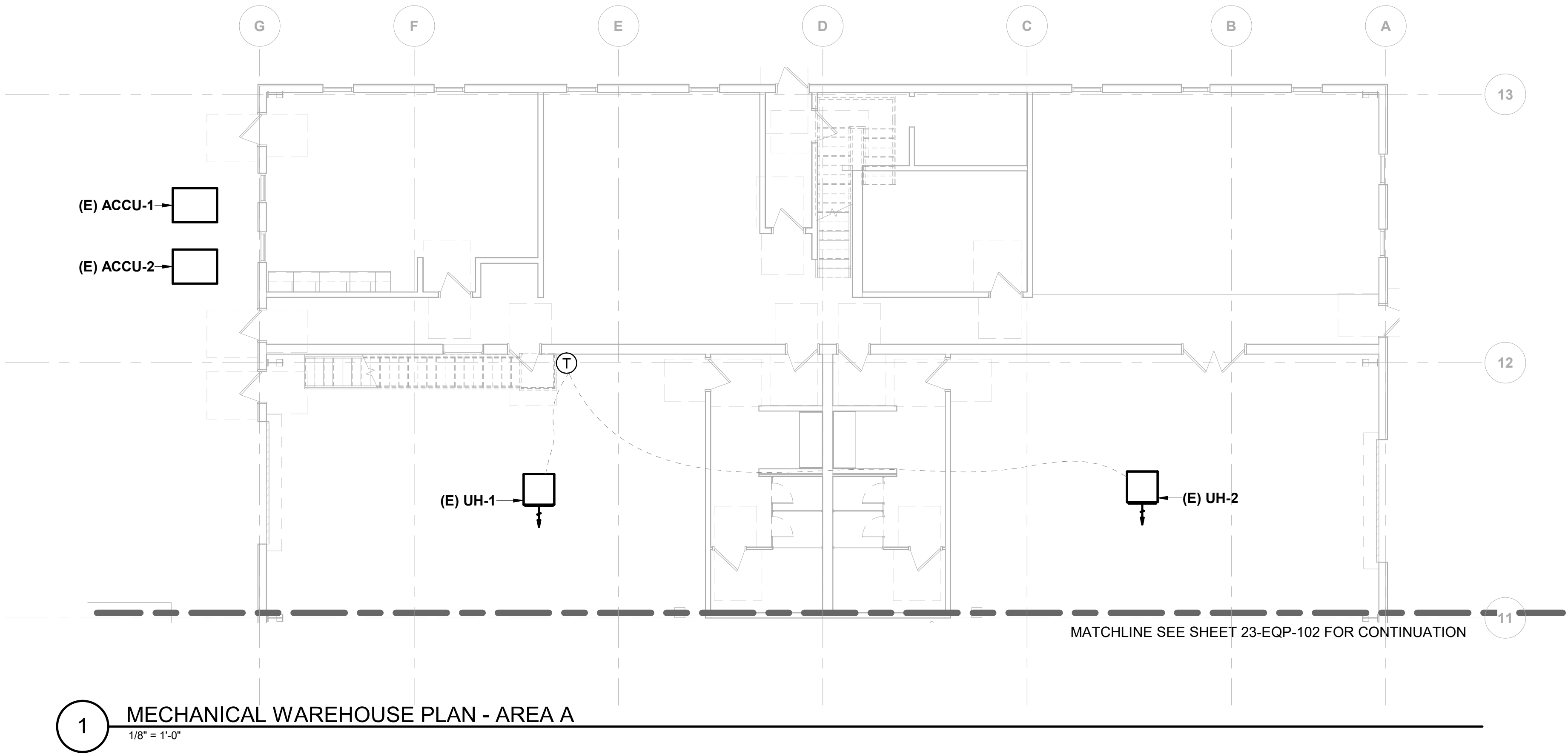
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APPENDIX B WAREHOUSE EQUIPMENT PLANS

FILE PATH: Autodesk Docs://147053_SPIRITIAHHANG/147053-Mech-Warehouse.rvt

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IAH WAREHOUSE
RENOVATION

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SHEET NAME: WAREHOUSE MECHANICAL PLAN

SHEET No. 23-EQP-101 SCALE: 1/8" = 1'-0"

SHEET SIZE: 22"x34" ANSI-D





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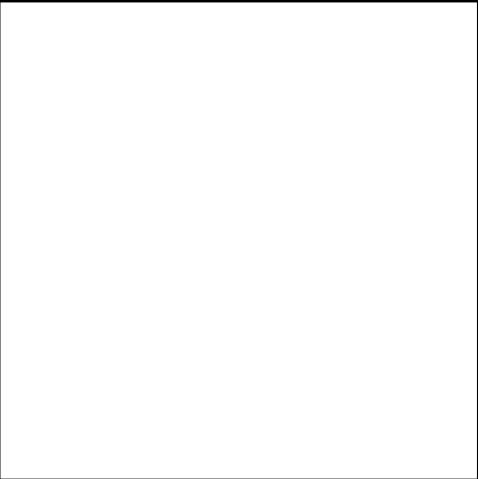
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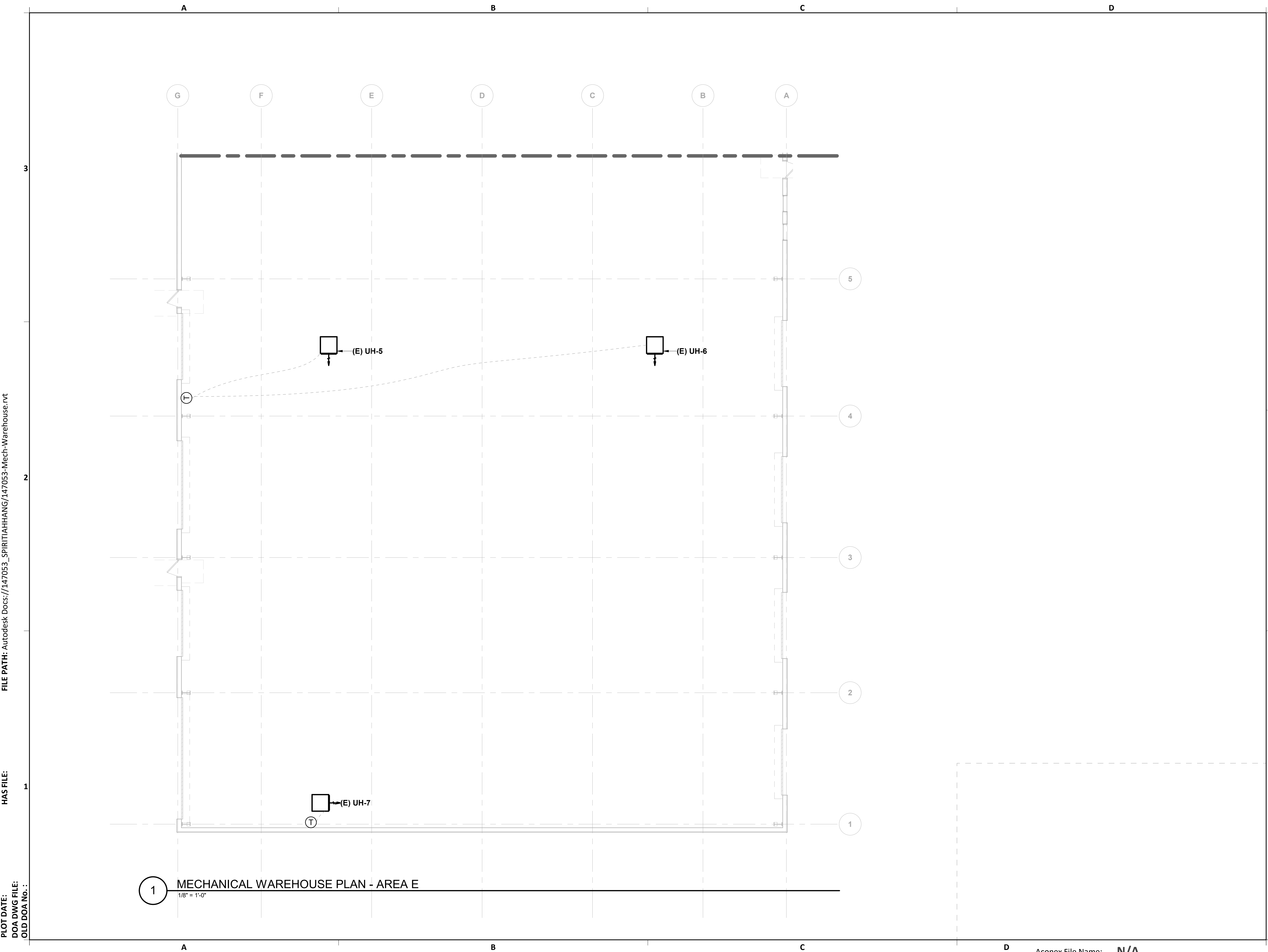
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SHEET NAME: WAREHOUSE MECHANICAL PLAN

SHEET No. 23-EQP-102	SCALE: 1/8" = 1'-0"
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SHEET SIZE: 22"x34" ANSI-D

Aconex File Name: **N/A**





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SHEET NAME: WAREHOUSE MECHANICAL PLAN

SHEET No. 23-EQP-103	SCALE: 1/8" = 1'-0"
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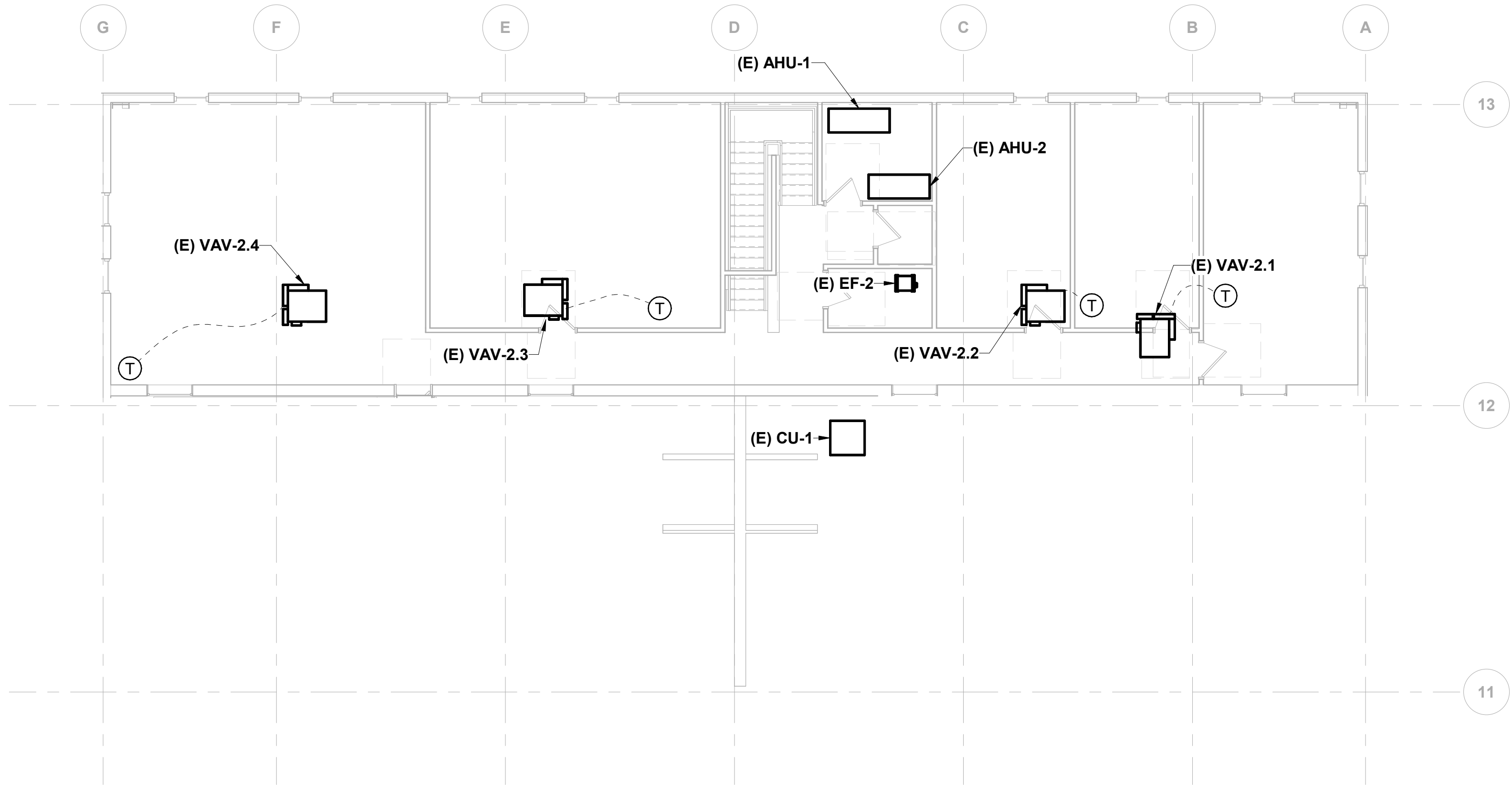
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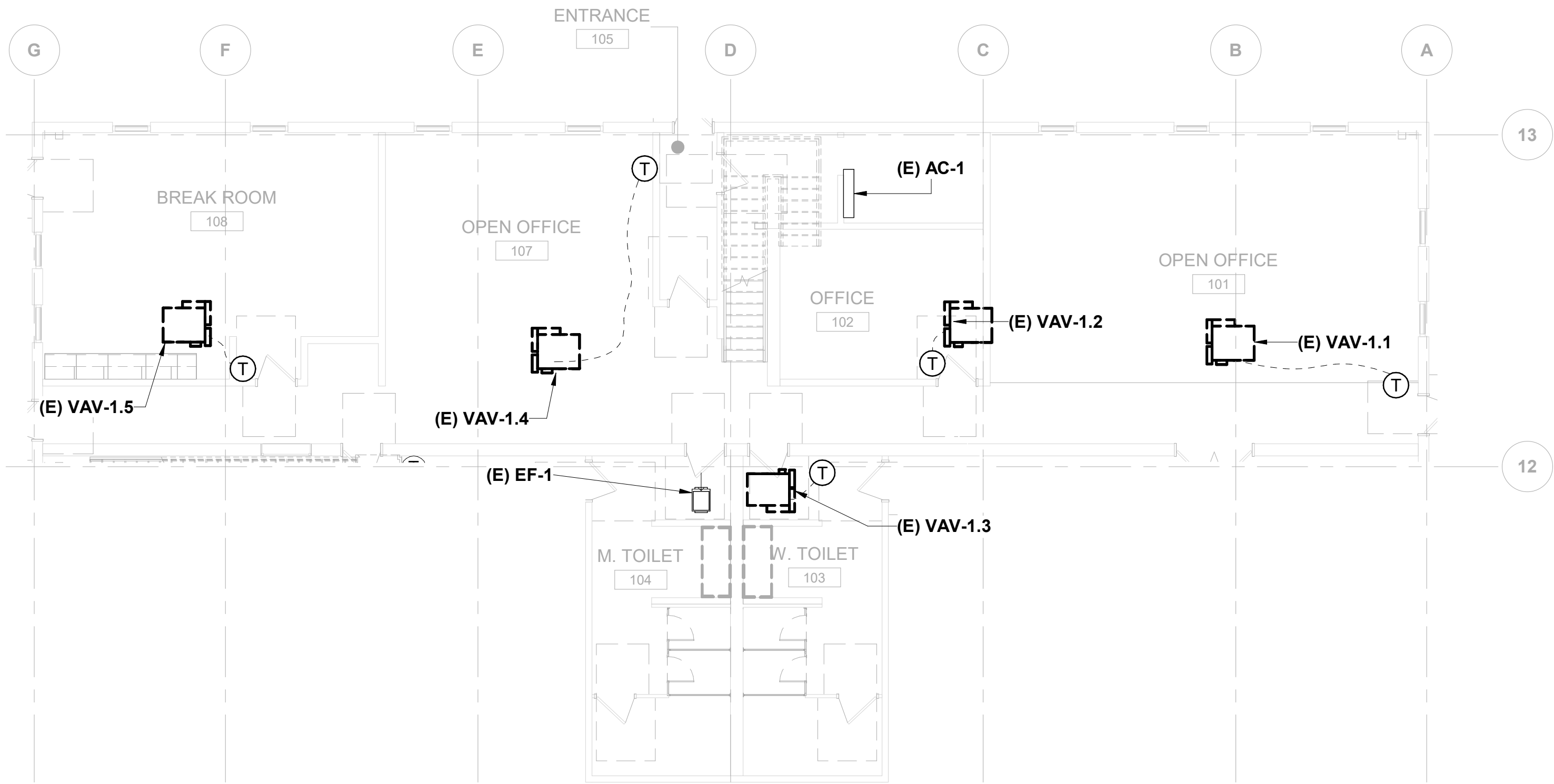
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PLOT DATE:
DOA DWG FILE:
OLD DOA No. :

2 WAREHOUSE ADMIN AREA - SECOND FLOOR MECHANICAL PLAN
1/8" = 1'-0"



1 WAREHOUSE ADMIN AREA - GROUND FLOOR MECHANICAL PLAN
1/8" = 1'-0"



SPIRIT AIRLINES
16825 JOHN F. KENNEDY BLVD.
HOUSTON, TX 77032

SPIRIT AIRLINES -
IAH WAREHOUSE
RENOVATION

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C.O.H. No.	N/A	D.O.A No.	N/A

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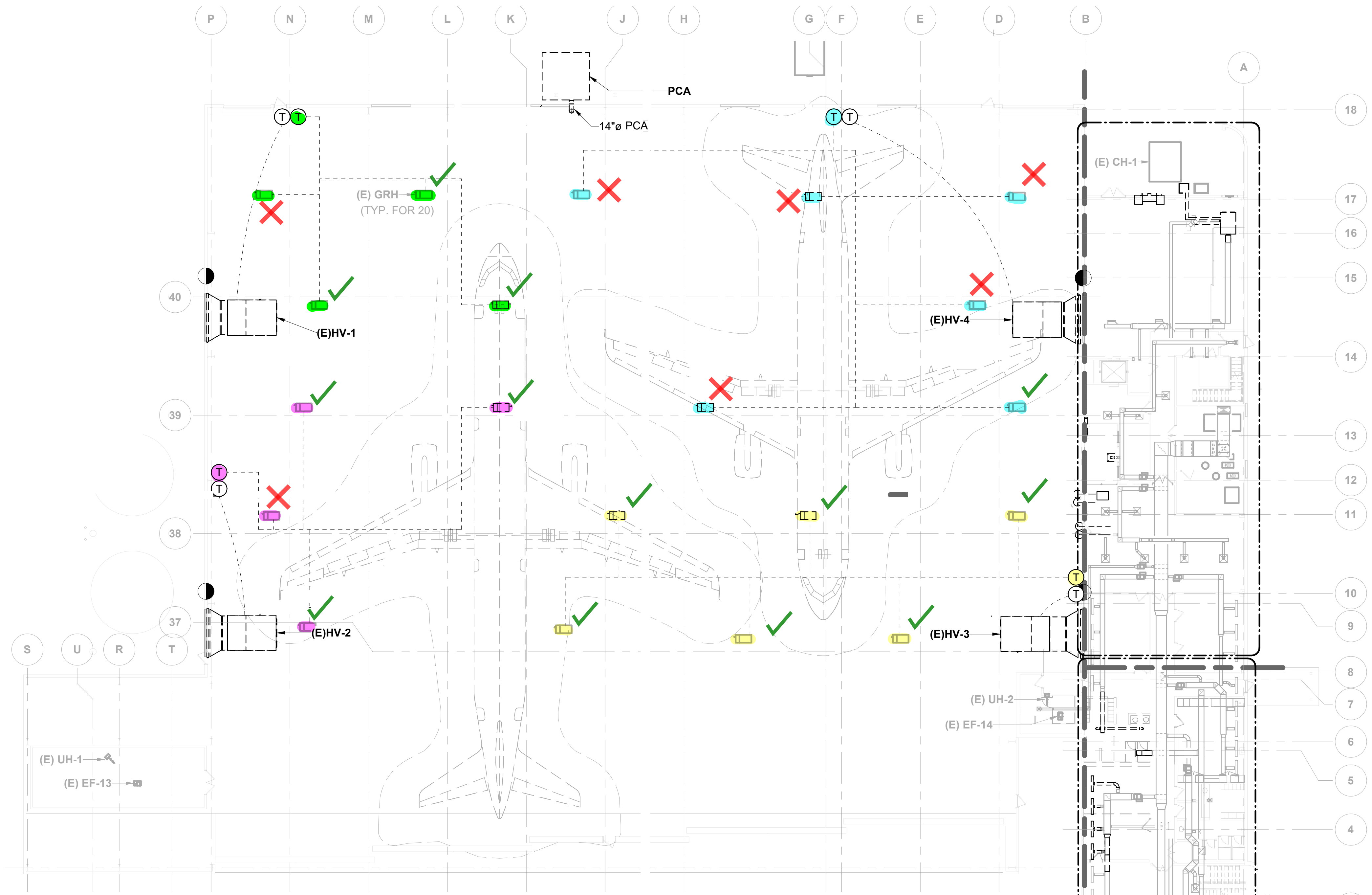
SHEET NAME: ADMIN AREA MECHANICAL PLANS

SHEET No. 23-EQP-104 SCALE: 1/8" = 1'-0"

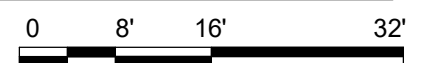
SHEET SIZE: 22"x34" ANSI-D

APPENDIX C HANGAR RADIANT HEATER ZONING PLAN

FILE PATH: Autodesk Docs://147053_SPIRITIAHANG/147053-Mech-Hangar.rvt
HAS FILE:
PLOT DATE:
DOA DWG FILE:
OLD DOA No.:



1 GROUND FLOOR - HANGAR GAS-RADIANT HEATER ZONING
1/16" = 1'-0"



GAS-RADIANT HEATER ZONING LEGEND	
	ZONE 1 - SOUTH-EAST ZONE
	ZONE 2 - NORTH-EAST ZONE
	ZONE 3 - NORTH-WEST ZONE
	ZONE 4 - SOUTH-WEST ZONE

- RADIANT HEATER ENABLED AND OPERATIONAL
- RADIANT HEATER DID NOT ENABLE

FOR REFERENCE
ONLY



SPIRIT AIRLINES
16875 JOHN F. KENNEDY BLVD.
HOUSTON, TX 77032

SPIRIT AIRLINES -
IAH HANGAR RENOVATION

T.I.P. No.	TIP-22-177-IAH	A.I.P. No.	N/A
C.O.H. No.	N/A	D.O.A No.	N/A



BURNS & MCDONNELL
BURNS & MCDONNELL ENGINEERING CO, INC
9400 WARD PARKWAY
KANSAS CITY, MO 64114
LICENSE NO. BR-2016
LICENSE NO. F-A45

DESIGNER PROJECT No.: 147053
PROJECT STATUS: CD

REVISIONS			
No.	DESCRIPTION	DATE	BY
0	ISSUED FOR PERMIT	10/14/22	AP

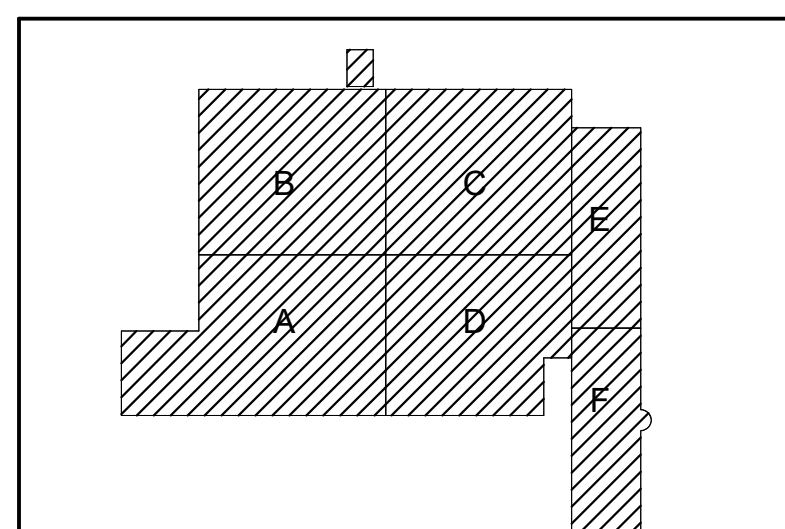
DESIGNER: A. PASQUALI
DRAWN BY: J. NGUYEN
CHECKED BY: A. THOMAS
ISSUE DATE: 10/14/22
APPROVED BY:
APPROVAL DATE: -

DIRECTOR
of
HOUSTON AIRPORT SYSTEM

Review/ Approval Category

IFC

ISSUED FOR CONSTRUCTION



KEY PLAN
NOT TO SCALE

SHEET NAME: GROUND FLOOR - MECHANICAL
OVERALL DEMO PLAN
SHEET No. MD101-H SCALE: 1/16" = 1'-0"

SHEET SIZE: 22"x34" ANSI-D

APPENDIX D RETRO-COMMISSIONING PLAN

SPIRIT AIRLINES IAH HANGAR RETROCOMMISSIONING PLAN

1. Commissioning Process Responsible Parties:

- CX – Commissioning Agent (BMcD)
- CM – Construction Manager – (CDS)
- MC – Mechanical Contractor (Lee Thompson)
- CC – Control Contractor (Lee Thompson)
- FM – Facilities Manager (Spirit Airlines)
- T&B – Testing and Balancing Specialist (T&B)

2. Pre-Functional Testing Activities

2.1 Control System Upgrade

Schedule: CM to coordinate with CC.

The existing Johnson Control (JCI) building automation system was inspected during the condition assessment and the system currently utilizes the Tridium AX Framework no longer supported by JCI. The system needs to be upgraded to the N4 platform. This includes replacing the existing FX60 supervisor controller with an FX80 controller, obtaining a new software license, and migrating the existing control sequences to the new equipment. All existing field (equipment) controllers to remain as installed. New FX80 Controller shall provide remote access capability for Spirit Airlines facilities maintenance personnel. CC shall document and coordinate with CC to document existing control programming and control points list.

Refer to the following for additional information:

- Lee Thompson Proposal - Upgrade Obsolete FX60 08.31.22. – (With BMcD Comments)
- BMcD IAH Spirit Airlines Condition Assessment

2.2 **Chilled Water System Repairs**

Schedule: CM to coordinate with MC

Engage a MC to perform required repairs observed during the site assessment. Repairs included the following:

- Flush chilled water system and treat loop with water treatment.
- Clean all strainers.
- Replace chilled water make-up water pressure regulating valve.
- Monitor refrigerant pressure within the chiller to determine identify is system it has significant leaking.
- Replace AHU 3-way chilled water control valve.
- Calibrate pressure and temperature gauges.

Upon completion of repairs, engage a T&B to perform system testing and balancing of the chilled water system. Balancing shall be performed with the new 3-way control valve in both the full bypass and full flow positions. Refer to testing and balancing activities for additional requirements.

Refer to the following documents for additional information:

- Lee Thompson Proposal – Chilled Water System Repairs 09.15.22
- BMcD IAH Spirit Airlines Condition Assessment

2.3 **Testing and Balancing**

Schedule: CM to coordinate with T&B

Engage a T&B to perform the following activities:

- AHU-1 supply air duct airflow leakage investigation.
 - o Supply air duct leakage was observed during the site assessment. Specialist to perform measurements to identify all locations required for repair. Perform prior to system testing and balancing.
- AHU-1 supply air distribution system testing and balancing.
 - o Refer to Testing and Balancing – Scope of work drawings for additional information.
- Perform testing & balancing of hot water heating system.

- o Refer to Testing and Balancing – Scope of work drawings for additional information.
- Perform testing and balancing of the chilled water system
 - o Note this testing and balancing is required as part of chilled system repair activities. Coordinate with chilled system repair activities.
 - o Refer to Testing and Balancing – Scope of work drawings for additional information.

2.4 Functional Testing Procedures Development (BMcD)

Schedule: CX to complete by 10/11.

CX to develop functional testing procedures for all existing systems and equipment indicated in the Existing System Commissioning List. Procedures will be based on the following information:

- Existing equipment and control system documentation.
- Coordination with CC performing the control system upgrade to identify existing control points and sequences.

3. Commissioning Activities

3.1 Functional Testing of Existing Equipment

Schedule: BMcD to schedule based on expected completion date of Pre-Commissioning Activities (Part 2) provided by CM.

- ▶ CX to execute functional testing of all existing systems and equipment indicated in the Existing System Commissioning List. Functional testing shall not begin until all pre-commissioning activities are substantially complete. Testing shall be coordinated and required support from the MC, CC, T&B, FM and CM. The functional test procedures are to be executed by the Contractors and witnessed by the CX. The CX will maintain log of issues found during commissioning & work with construction team to find resolutions to issues. Once issues are resolved, CX will verify resolution.

4. Existing System Commissioning List

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) CH-1	CHILLER	HANGAR ADMIN. AREA
HANGAR	(E) CHWP-1	PUMP	(E) CH-1
HANGAR	(E) CHWP-2	PUMP	(E) CH-1

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) B-1	BOILER	HANGAR ADMIN. AREA
HANGAR	(E) HWP-1	PUMP	(E) B-1

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) AHU-1	AIR-HANDLING UNIT	HANGAR ADMIN. AREA

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) VAV-01	FAN-POWERED BOX	LOBBY 108
HANGAR	(E) VAV-02	FAN-POWERED BOX	SECURITY 109
HANGAR	(E) VAV-03	FAN-POWERED BOX	OFFICE 112
HANGAR	(E) VAV-04	FAN-POWERED BOX	KITCHEN 118
HANGAR	(E) VAV-05	FAN-POWERED BOX	ADMIN GROUND FLR CORRIDOR
HANGAR	(E) VAV-06	FAN-POWERED BOX	SHOP 128
HANGAR	(E) VAV-07	FAN-POWERED BOX	SHOP 128
HANGAR	(E) VAV-08	FAN-POWERED BOX	SHOP 128
HANGAR	(E) VAV-09	FAN-POWERED BOX	ADMIN GROUND FLR CORRIDOR
HANGAR	(E) VAV-10	FAN-POWERED BOX	OFFICE 202
HANGAR	(E) VAV-11	FAN-POWERED BOX	OFFICE 202
HANGAR	(E) VAV-12	FAN-POWERED BOX	OFFICE 202
HANGAR	(E) VAV-13	FAN-POWERED BOX	OFFICE 202
HANGAR	(E) VAV-14	FAN-POWERED BOX	OFFICE 208
HANGAR	(E) VAV-15	FAN-POWERED BOX	FLIGHT OPERATIONS 217
HANGAR	(E) VAV-16	FAN-POWERED BOX	ACCOUNTING 216
HANGAR	(E) VAV-17	FAN-POWERED BOX	COFFEE 219
HANGAR	(E) VAV-18	FAN-POWERED BOX	CONF RM 218
HANGAR	(E) VAV-19	FAN-POWERED BOX	TRAINING DEVELOPMENT 224
HANGAR	(E) VAV-20	FAN-POWERED BOX	PILOTS OFFICE
HANGAR	(E) VAV-21	FAN-POWERED BOX	OFFICE 230
HANGAR	(E) VAV-22	FAN-POWERED BOX	IT RM 231
HANGAR	(E) VAV-23	VAV BOX	SHOP 128

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) CRAC-1	COMPUTER ROOM UNIT	IT RM 231
HANGAR	(E) CRAC-2	COMPUTER ROOM UNIT	IT RM 231
HANGAR	(E) AHU-2	AIR-HANDLING UNIT	IT RM 231
HANGAR	(E) CU-5	CONDENSING UNIT	(E) CRAC-2
HANGAR	(E) CU-6	CONDENSING UNIT	(E) CRAC-1

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) GRH	GAS RADIANT HEATER	HANGAR

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) HV-6	HEATING VENTILATION UNIT	STORAGE 235
HANGAR	(E) CU-2	CONDENSING UNIT	HV-6
HANGAR	(E) EF-10	EXHAUST FAN	STORAGE 235

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) UH-1	UNIT HEATER	PUMP RM 100B
HANGAR	(E) EF-13	EXHAUST FAN	PUMP RM 100B

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) UH-2	UNIT HEATER	HANGAR RESTROOM 100A
HANGAR	(E) EF-14	EXHAUST FAN	HANGAR RESTROOM 100A

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) EF-08	EXHAUST FAN	ELEC. RM 233

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
HANGAR	(E) EF-09	EXHAUST FAN	RM 133 & RM 132

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
WAREHOUSE	(E) UH-1-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-2-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-3-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-4-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-5-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-6-W	UNIT HEATER	100 WAREHOUSE
WAREHOUSE	(E) UH-7-W	UNIT HEATER	100 WAREHOUSE

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
WAREHOUSE	(E) AC-1-W	MINI-SPLIT AC	111 COMM.
WAREHOUSE	(E) CU-1-W	CONDENSING UNIT	(E)AC-1-2



BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
WAREHOUSE	(E) EF-1-W	EXHAUST FAN	1ST FL. RESTROOMS

BUILDING	UNIT	EQUIPMENT TYPE	SERVICE
WAREHOUSE	(E) EF-2-W	EXHAUST FAN	2ND FL. RESTROOM



APPENDIX E FUNCTIONAL PERFORMANCE TESTING

Spirit Airlines
IAH - Warehouse and Hanger

(E) AHU-1 (RCx - AHU-1)
FPT-1

FPT-1: (E) AHU-1 (RCx - AHU-1)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) AHU-1
Type	AIR HANDLING UNIT
Description	AIR HANDLING UNIT
Discipline	HVAC
Location(s)	HANGAR>MECHANICAL RM 130



Questionnaire

#	Question	Answer	Details
Point-to-Point			

Inputs

1	OA Damper Position	☐ Pass ☒ Fail ☐ N/A	No feedback. Currently operates as when OA temp is above 50F, OA damper opens. OA damper actuator is not functioning.
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Checked by: Zach Yohon on 03/16/2023 01:58 PM EDT



Spirit Airlines
IAH - Warehouse and Hanger

(E) AHU-1 (RCx - AHU-1)
FPT-1

#	Question	Answer	Details
2	Supply Fan Status	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
3	Supply Fan Speed	☐ Pass ☒ Fail ☐ N/A	No feedback on VFD speed Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
4	Discharge Air Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
5	Return Air Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
6	Relative Humidity	☐ Pass ☐ Fail ☒ N/A	No RH sensors Checked by: Zach Yohon on 02/23/2023 12:34 PM EST
7	CHW Valve Status	☐ Pass ☒ Fail ☐ N/A	Currently no valve status. However, chilled water valve does have feedback capabilities. Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
8	CHWP-1 Status	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
9	CHWP-2 Status	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
Outputs			
1	Supply Fan Command	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
2	OA Damper Command	☒ Pass ☐ Fail ☐ N/A	No feedback. Currently operates as when OA temp is above 50F, OA damper opens. OA damper actuator is not functioning. Checked by: Zach Yohon on 03/16/2023 01:58 PM EDT
3	CHW Valve Command	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
4	CHWP-1 Command	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 11:16 AM EST
5	CHWP-2 Command	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 11:16 AM EST

#	Question	Answer	Details
Sequence of Operations			
Static Pressure Control			
1	Simulate a zone damper position exceeding 85% open -Verify AHU ramps supply fan speed to raise duct static pressure	☒ Pass ☐ Fail ☐ N/A	
Checked by: Zach Yohon on 02/23/2023 11:59 AM EST			
2	Simulate air flow through a VAV to be 30% or more below the setpoint for 15 minutes -Verify AHU ramps up supply fan speed further to raise duct static pressure -After the air flow is within 10% of the setpoint, verify AHU ramps down supply fan speed to lower duct static pressure	☒ Pass ☐ Fail ☐ N/A	Static pressure sensor located on the SA main to the first floor only. No static pressure sensor in the 2nd floor SA main. SF VFD modulates based on 1st floor SA main.
Checked by: Zach Yohon on 02/23/2023 12:08 PM EST			
3	Simulate the above zone damper position closing below 80% open -Verify AHU ramps supply fan speed to lower duct static pressure	☒ Pass ☐ Fail ☐ N/A	
Checked by: Zach Yohon on 02/23/2023 11:59 AM EST			
4	Simulate the relief air static pressure below the setpoint -Verify relief fan ramps up to raise the relief air static pressure	☐ Pass ☐ Fail ☒ N/A	No relief air for AHU-1
Checked by: Zach Yohon on 02/23/2023 11:49 AM EST			
5	Simulate the relief air static pressure below the setpoint -Verify relief fan ramps up to raise the relief air static pressure	☐ Pass ☐ Fail ☒ N/A	No relief air for AHU-1
Checked by: Zach Yohon on 02/23/2023 11:49 AM EST			
6	Simulate the mixed air plenum static pressure below the setpoint (not in economizer mode) -Verify return air damper modulates open to raise the mixed air plenum static pressure	☐ Pass ☐ Fail ☒ N/A	Return air dampers are manual. No motorized dampers in return air duct.
Checked by: Zach Yohon on 02/23/2023 11:57 AM EST			
7	Simulate the mixed air plenum static pressure above the setpoint (not in economizer mode) -Verify return air damper modulates shut to lower the mixed air plenum static pressure	☐ Pass ☐ Fail ☒ N/A	Return air dampers are manual. No motorized dampers in return air duct.
Checked by: Zach Yohon on 02/23/2023 11:57 AM EST			
Relief Air Damper			

Spirit Airlines
IAH - Warehouse and Hanger

(E) AHU-1 (RCx - AHU-1)
FPT-1

#	Question	Answer	Details
1	Simulate TT-1 building static pressure below the setpoint -Verify AHU relief damper (D-2) modulates closed to raise the building static pressure	☐ Pass ☐ Fail ☒ N/A	No relief air for AHU-1
Checked by: Zach Yohon on 02/23/2023 11:49 AM EST			
2	Simulate TT-2 building static pressure above the setpoint -Verify AHU relief damper modulates open to lower the building static pressure	☐ Pass ☐ Fail ☒ N/A	No relief air for AHU-1
Checked by: Zach Yohon on 02/23/2023 11:49 AM EST			
Cooling			
1	Simulate OA temp above the cooling setpoint to place the AHU into cooling mode -Verify chilled water flow modulates to maintain supply air temp at the setpoint	☒ Pass ☐ Fail ☐ N/A	
Checked by: Zach Yohon on 02/23/2023 12:08 PM EST			
2	Simulate OA temp further above the cooling setpoint -Verify chilled water flow modulates to maintain supply air temp at the setpoint	☒ Pass ☐ Fail ☐ N/A	
Checked by: Zach Yohon on 02/23/2023 12:08 PM EST			
3	Simulate OA temp below the heating setpoint to place the AHU into heating mode -Verify chilled water flow modulates to maintain supply air temp at the setpoint	☒ Pass ☐ Fail ☐ N/A	AHU-1 does not have heating. CHW valve modulates correctly to maintain
Checked by: Zach Yohon on 02/23/2023 12:29 PM EST			
4	Simulate OA temp further below the heating setpoint -Verify chilled water valve modulates closed to maintain supply air temp at the setpoint	☒ Pass ☐ Fail ☐ N/A	AHU-1 does not have heating. CHW valve modulates correctly to maintain
Checked by: Zach Yohon on 02/23/2023 12:29 PM EST			
Discharge Air Temp Reset			
1	Simulate the relative humidity below 50%. -Discharge air temperature shall be increased to 55°F.	☐ Pass ☐ Fail ☒ N/A	System does not have RH sensor
Checked by: Zach Yohon on 02/23/2023 12:10 PM EST			
2	Simulate the relative humidity above 70%. -Discharge air temperature shall be decreased to 45°F.	☐ Pass ☐ Fail ☒ N/A	System does not have RH sensor
Checked by: Zach Yohon on 02/23/2023 12:10 PM EST			
Economizer			

#	Question	Answer	Details
1	Simulate OA enthalpy less than RA enthalpy and OA temperature above 52°F. Verify the following: -OA damper fully opens -RA damper fully closes -Cooling valve modulates to maintain discharge air temperature setpoint	☐ Pass ☐ Fail ☒ N/A	AHU-1 does not have economizer cycle
Checked by: Zach Yohon on 02/23/2023 12:08 PM EST			
2	Simulate OA enthalpy less than RA enthalpy and OA temperature below 52°F -Verify the OA damper and RA damper modulate to maintain the mixed air temperature at the setpoint	☐ Pass ☐ Fail ☒ N/A	AHU-1 does not have economizer cycle
Checked by: Zach Yohon on 02/23/2023 12:08 PM EST			
3	Simulate the mixed air temp below 45°F. Verify the following: -Minimum outdoor air damper returns to minimum position -Economizer outdoor air damper fully closes -RA damper modulates open	☐ Pass ☐ Fail ☒ N/A	AHU-1 does not have economizer cycle
Checked by: Zach Yohon on 02/23/2023 12:08 PM EST			
Alarms and Interlocks			
1	Simulate supply duct static pressure above the setpoint -Verify a High Duct Air Pressure alarm is generated	☐ Pass ☒ Fail ☐ N/A	No alarm currently programmed
Checked by: Zach Yohon on 02/23/2023 12:34 PM EST			
2	Simulate a loss of supply fan -Verify a Supply Fan Not Proven alarm is generated	☐ Pass ☒ Fail ☐ N/A	No alarm currently programmed
Checked by: Zach Yohon on 02/23/2023 12:34 PM EST			
3	Simulate a loss of return fan -Verify a Return Fan Not Proven alarm is generated	☐ Pass ☐ Fail ☒ N/A	System does not utilize return fan
Checked by: Zach Yohon on 02/23/2023 12:34 PM EST			
4	Simulate mixed air temperature below the setpoint -Verify a Low Mixed Air Temperature alarm is generated	☐ Pass ☐ Fail ☒ N/A	No mixed air temperature sensor
Checked by: Zach Yohon on 02/23/2023 12:34 PM EST			
5	Simulate supply air temperature above the setpoint in excess of 3°F -Verify a High Supply Air Temperature alarm is generated	☐ Pass ☒ Fail ☐ N/A	No alarm currently programmed
Checked by: Zach Yohon on 02/23/2023 12:34 PM EST			

#	Question	Answer	Details
6	Simulate supply air temperature below the setpoint in excess of 3°F -Verify a Low Supply Air Temperature alarm is generated	☐ Pass ☒ Fail ☐ N/A	No alarm currently programmed
Checked by: Zach Yohon on 02/23/2023 12:34 PM EST			
7	Simulate smoke detected at SD-1 (1st Floor RA) -Verify Unit Shuts Down and Alarm is sent to BMS	☐ Pass ☐ Fail ☒ N/A	Could not be tested while on-site.
Checked by: Zach Yohon on 03/20/2023 11:39 AM EDT			
8	Simulate smoke detected at SD-2 (2nd Floor RA) -Verify Unit Shuts Down and Alarm is sent to BMS	☐ Pass ☐ Fail ☒ N/A	Could not be tested while on-site.
Checked by: Zach Yohon on 03/20/2023 11:39 AM EDT			
9	Simulate smoke detected at SD-3 (SA) -Verify Unit Shuts Down and Alarm is sent to BMS	☐ Pass ☐ Fail ☒ N/A	Could not be tested while on-site.
Checked by: Zach Yohon on 03/20/2023 11:39 AM EDT			

Related Issues

#: 27	Asset: (E) AHU-1	Code: FPT-1-001	Status: Open	Priority: Low
Reported By: Zach Yohon on 02/23/2023				
Location(s): HANGAR>MECHANICAL RM 130				
Assigned to : Unassigned				
Description: AHU-1 outdoor air damper does not operate to maintain required outdoor airflow required per space/building. OA damper actuator is not functioning and stays in open position. No airflow measuring station on outdoor air duct. Currently controls calls for OA damper to open when outdoor air is above 50 deg F and close when OA is below 50 deg F.				
Response(s): None				

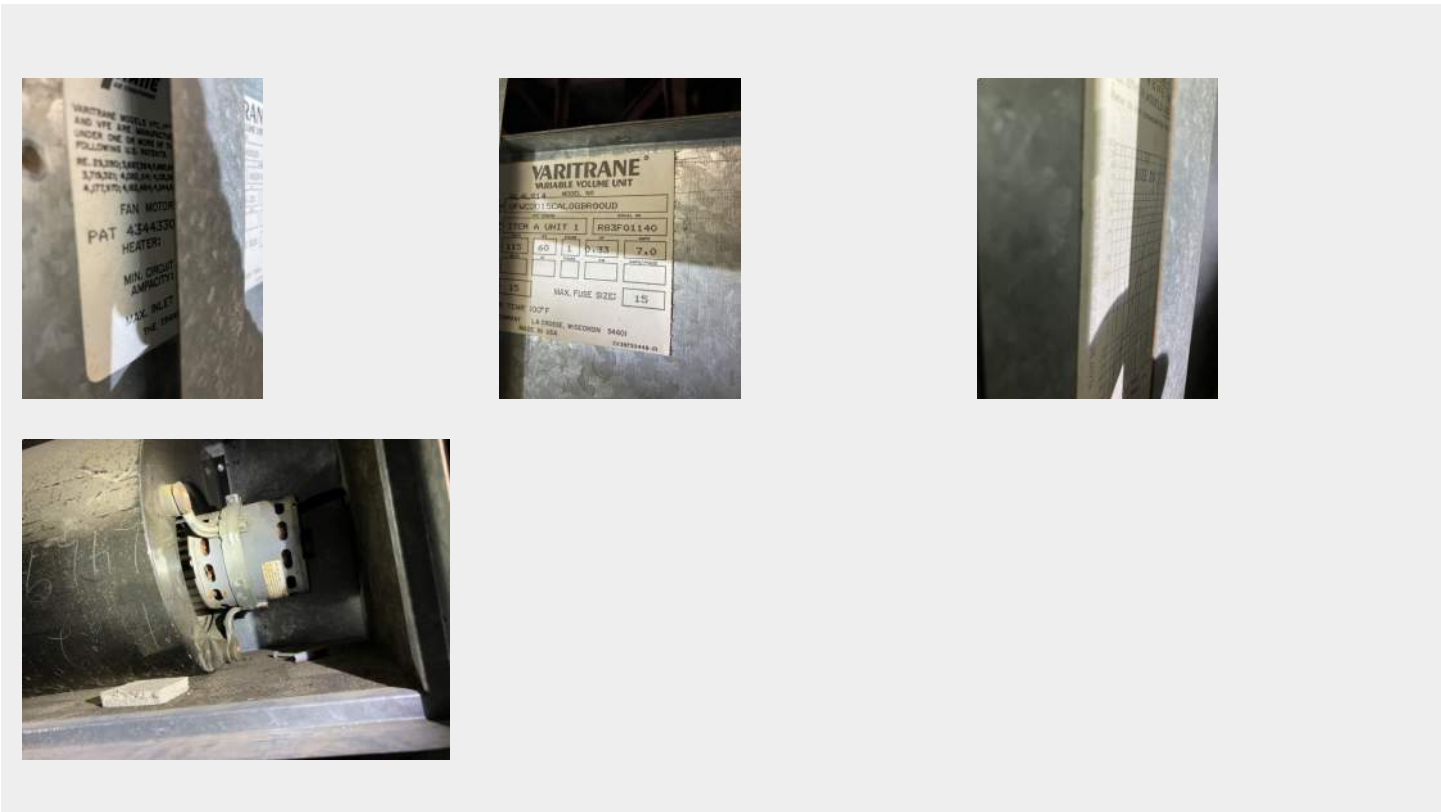
END TEST

FPT-2: (E) VAV-1 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary	
Unit #	(E) VAV-1
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>LOBBY 108



Set Points

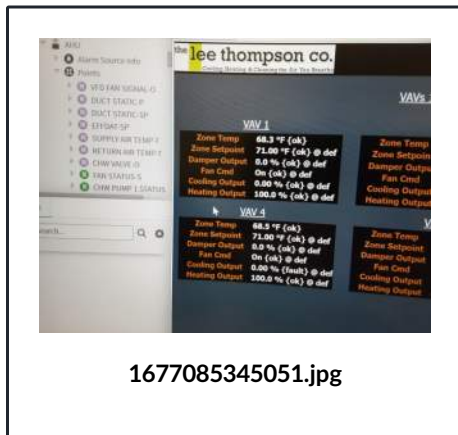
Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

1 Room Temperature ☒ Pass ☐ Fail ☐ N/A



Checked by: Theo Lagonikos on 02/22/2023 11:02 AM EST

#	Question	Answer	Details
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Theo Lagonikos on 02/22/2023 11:03 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only BMS command to valve Checked by: Zach Yohon on 03/16/2023 02:00 PM EDT
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Theo Lagonikos on 02/22/2023 11:38 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Theo Lagonikos on 02/22/2023 11:38 AM EST
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 02/28/2023 11:15 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	 Checked by: Theo Lagonikos on 02/22/2023 11:38 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	 Checked by: Theo Lagonikos on 02/22/2023 11:38 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Can keep as is or reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	 Checked by: Theo Lagonikos on 02/22/2023 11:36 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	 Checked by: Theo Lagonikos on 02/22/2023 11:36 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Theo Lagonikos on 02/22/2023 11:54 AM EST

Occupied Mode

#	Question	Answer	Details
1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 11:43 AM EST			
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling
Checked by: Theo Lagonikos on 02/22/2023 03:46 PM EST			
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed
Checked by: Theo Lagonikos on 02/22/2023 12:11 PM EST			
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 12:08 PM EST			
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow.
Checked by: Theo Lagonikos on 02/22/2023 02:43 PM EST			
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 12:24 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 11:48 AM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 02:06 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 12:07 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 02:09 PM EST			
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	87F
Checked by: Theo Lagonikos on 02/23/2023 10:51 AM EST			
Alarms and Safeties			

#	Question	Answer	Details
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Theo Lagonikos on 02/22/2023 12:15 PM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Theo Lagonikos on 02/22/2023 12:23 PM EST

END TEST

FPT-3: (E) VAV-2 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-2
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>SECURITY 109



Set Points		Setpoint		Alarm	
Parameter	Description / Trend Information	Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:29 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 10:29 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 10:29 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:29 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:29 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:29 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:29 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:29 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Can keep as is or reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:29 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:29 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:29 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 01:53 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables Checked by: Theo Lagonikos on 02/22/2023 02:45 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 02:26 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 02:26 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow. Checked by: Theo Lagonikos on 02/22/2023 02:11 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 02:32 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 01:56 PM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 02:05 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 02:25 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 02:09 PM EST			
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	99F
Checked by: Theo Lagonikos on 02/23/2023 10:51 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:30 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:30 AM EST			

END TEST

FPT-4: (E) VAV-3 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-3
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>OFFICE 112



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:32 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/06/2023 10:32 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve
Checked by: Zach Yohon on 03/06/2023 10:32 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:32 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:32 AM EST
- Heating temperature setpoint ☒ Pass ☐ Fail ☐ N/A Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint
Checked by: Zach Yohon on 03/06/2023 10:32 AM EST
- Min Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:32 AM EST
- Max Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:32 AM EST

#	Question	Answer	Details
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Can keep as is or reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:32 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:32 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:32 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 02:33 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/23/2023 10:23 AM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 02:42 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 10:23 AM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow. Checked by: Theo Lagonikos on 02/22/2023 02:43 PM EST
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 10:49 AM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 02:37 PM EST
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#	Question	Answer	Details
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/22/2023 02:38 PM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 10:19 AM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 02:40 PM EST
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	88F Checked by: Theo Lagonikos on 02/23/2023 10:24 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 10:34 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 10:34 AM EST

END TEST

FPT-5: (E) VAV-4 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-4
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>KITCHEN 118



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve
Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
- Heating temperature setpoint ☒ Pass ☐ Fail ☐ N/A Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint
Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
- Min Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
- Max Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:36 AM EST

#	Question	Answer	Details
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Can keep as is or reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:36 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:05 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables Checked by: Theo Lagonikos on 02/22/2023 03:17 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 03:39 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:17 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 03:17 PM EST
5	Record discharge air temperature with temperature gun	☐ Pass ☒ Fail ☐ N/A	Could not verify 55-58 SAT Checked by: Theo Lagonikos on 02/22/2023 03:22 PM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:07 PM EST
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Spirit Airlines
IAH - Warehouse and Hanger

(E) VAV-4 (RCx VAV - Variable Air Volume Terminal ...
FPT-5

#	Question	Answer	Details
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/22/2023 03:09 PM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:17 PM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 03:09 PM EST
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	90F Checked by: Theo Lagonikos on 02/23/2023 10:52 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 10:36 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 10:36 AM EST

Related Issues

#: 29	Asset: (E) VAV-4	Code: FPT-5-001	Status: Open	Priority: Low
Reported By: Theo Lagonikos on 03/03/2023				
Location(s): HANGAR>KITCHEN 118				
Assigned to : Unassigned				
Description: VAV-4: Supply air temperature to zone reading above 68 deg F. AHU DAT reading approx. 53 deg F. Investigate airflow issues at VAV-4				
Response(s): None				

Spirit Airlines
IAH - Warehouse and Hanger

(E) VAV-4 (RCx VAV - Variable Air Volume Terminal ...
FPT-5

FPT-6: (E) VAV-5 (RCx VAV - Variable Air Volume Terminal Unit)

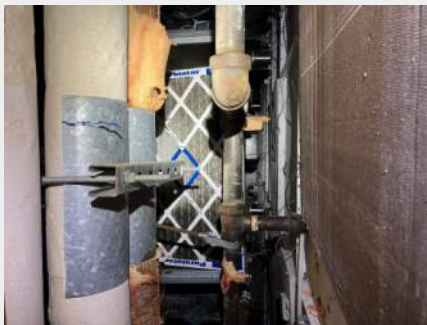
END TEST

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-5
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>ADMIN GROUND FLR CORRIDOR



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve
Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
- Heating temperature setpoint ☒ Pass ☐ Fail ☐ N/A Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint
Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
- Min Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
- Max Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:37 AM EST

#	Question	Answer	Details
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Can keep as is or reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:37 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:23 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables Checked by: Theo Lagonikos on 02/22/2023 03:28 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 03:39 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:28 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 03:29 PM EST
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:33 PM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:27 PM EST
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#	Question	Answer	Details
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/22/2023 03:23 PM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:29 PM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 03:24 PM EST
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	95F Checked by: Theo Lagonikos on 02/23/2023 10:52 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 10:37 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 10:37 AM EST

END TEST

FPT-7: (E) VAV-6 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-6
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>SHOP 128



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve
Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
- Heating temperature setpoint ☒ Pass ☐ Fail ☐ N/A Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint
Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
- Min Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
- Max Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:40 AM EST

#	Question	Answer	Details
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Can keep as is or reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:40 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:34 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan is disabled during cooling Checked by: Zach Yohon on 02/28/2023 11:24 AM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 03:42 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:42 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 03:35 PM EST
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:40 PM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☐ Pass ☒ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:47 PM EST
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#	Question	Answer	Details
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/22/2023 03:34 PM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:44 PM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 03:34 PM EST
5	Record discharge air temperature	☐ Pass ☒ Fail ☐ N/A	70F Checked by: Theo Lagonikos on 02/23/2023 10:53 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 10:40 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 10:40 AM EST

Related Issues

#: 17	Asset: (E) VAV-6	Code: FPT-7-001	Status: Open	Priority: Low
Reported By: Theo Lagonikos on 02/22/2023				
Assigned to : Unassigned				
Description: VAV fan doesn't energize				
Response(s): None				

END TEST

FPT-8: (E) VAV-7 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-7
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>SHOP 128



Set Points					
Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:44 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:44 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:44 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:48 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan is disabled during cooling Checked by: Theo Lagonikos on 02/22/2023 04:01 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 04:00 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 03:58 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 03:59 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 03:53 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☐ Pass ☒ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 03:56 PM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 03:59 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☐ Pass ☒ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 03:57 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 03:59 PM EST			
5	Record discharge air temperature	☐ Pass ☒ Fail ☐ N/A	68F
Checked by: Theo Lagonikos on 02/23/2023 10:54 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:44 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:44 AM EST			

Spirit Airlines
IAH - Warehouse and Hanger

(E) VAV-7 (RCx VAV - Variable Air Volume Terminal ...
FPT-8

Related Issues

#: 18 Asset: (E) VAV-7 Code: FPT-8-001 Status: Open Priority: Low

Reported By: Theo Lagonikos on 02/22/2023

Assigned to : Unassigned

Description:
VAV fan did not energize

Response(s):
None

#: 19 Asset: (E) VAV-7 Code: FPT-8-002 Status: Open Priority: Low

Reported By: Theo Lagonikos on 02/22/2023

Assigned to : Unassigned

Description:
Valve not functioning

Response(s):
None

END TEST

FPT-9: (E) VAV-8 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-8
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>SHOP 128



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve
Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
- Heating temperature setpoint ☒ Pass ☐ Fail ☐ N/A Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint
Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
- Min Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
- Max Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:45 AM EST

#	Question	Answer	Details
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:45 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:06 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 04:10 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 04:09 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:08 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 04:08 PM EST
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:07 PM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:09 PM EST
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#	Question	Answer	Details
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/22/2023 04:09 PM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:09 PM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 04:09 PM EST
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	90F Checked by: Theo Lagonikos on 02/23/2023 10:54 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 10:45 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 10:45 AM EST

END TEST

FPT-10: (E) VAV-9 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-9
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>ADMIN GROUND FLR CORRIDOR



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A

Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring

Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve

Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor

Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor

Checked by: Zach Yohon on 03/06/2023 10:46 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:46 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:46 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:01 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 04:03 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 04:04 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:04 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 04:04 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:02 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:04 PM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 04:05 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:04 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 04:05 PM EST			
5	Record discharge air temperature	☐ Pass ☒ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/23/2023 10:54 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:46 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:46 AM EST			

Spirit Airlines
IAH - Warehouse and Hanger

(E) VAV-9 (RCx VAV - Variable Air Volume Terminal ...
FPT-10

Related Issues

#: 26 Asset: (E) VAV-9 Code: FPT-10-001 Status: Open Priority: Low

Reported By: Theo Lagonikos on 02/23/2023

Assigned to : Unassigned

Description:
Discharge air temperature is not rising above 72F.

Response(s):
None

END TEST

FPT-11: (E) VAV-10 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-10
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>OFFICE 202



Set Points		Setpoint		Alarm	
Parameter	Description / Trend Information	Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:47 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:47 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:47 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:12 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 04:16 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 04:16 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:16 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 04:16 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:15 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/23/2023 10:08 AM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 04:14 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:17 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 04:14 PM EST			
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	97F
Checked by: Theo Lagonikos on 02/23/2023 10:08 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:47 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:47 AM EST			

END TEST

FPT-12: (E) VAV-11 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-11
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>OFFICE 202



Set Points		Setpoint		Alarm	
Parameter	Description / Trend Information	Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:49 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:49 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:49 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:19 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 04:20 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/23/2023 10:11 AM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 10:11 AM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 04:28 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:21 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:30 PM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 04:19 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/23/2023 10:11 AM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 04:19 PM EST			
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	92F
Checked by: Theo Lagonikos on 02/23/2023 10:11 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:49 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:49 AM EST			

END TEST

FPT-13: (E) VAV-12 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-12
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>OFFICE 202



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A

Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring

Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve

Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor

Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor

Checked by: Zach Yohon on 03/06/2023 10:52 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Can keep as is or reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:52 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:52 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:31 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 04:39 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 04:40 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:40 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 04:39 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:39 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/23/2023 10:12 AM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 04:41 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:41 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 04:41 PM EST			
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	94F
Checked by: Theo Lagonikos on 02/23/2023 10:12 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:52 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:52 AM EST			

END TEST

FPT-14: (E) VAV-13 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-13
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>OFFICE 202



Set Points		Setpoint		Alarm	
Parameter	Description / Trend Information	Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:54 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:54 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:45 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 04:47 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/22/2023 04:48 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 04:48 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 04:48 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:48 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/23/2023 10:13 AM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 04:45 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 04:49 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 05:08 PM EST			
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	94F
Checked by: Theo Lagonikos on 02/23/2023 10:13 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:54 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:54 AM EST			

END TEST

Set Points		Setpoint		Alarm	
Parameter	Description / Trend Information	Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:54 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:54 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:54 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 10:43 AM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/23/2023 03:11 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/23/2023 03:11 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 03:11 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/23/2023 03:11 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/23/2023 03:13 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/23/2023 03:08 PM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/23/2023 03:08 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/23/2023 03:08 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/23/2023 03:08 PM EST			
5	Record discharge air temperature	☐ Pass ☐ Fail ☒ N/A	No hot water from boiler during testing
Checked by: Zach Yohon on 03/06/2023 10:55 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:54 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:54 AM EST			

END TEST

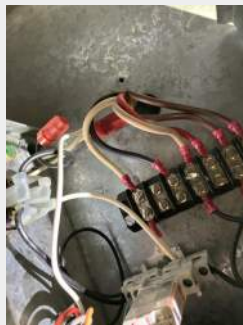
FPT-16: (E) VAV-15 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-15
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>FLIGHT OPERATIONS 217



Set Points		Setpoint		Alarm	
Parameter	Description / Trend Information	Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:56 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:56 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:12 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan is disabled during cooling Checked by: Theo Lagonikos on 02/22/2023 05:22 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 05:22 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:22 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 05:23 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☐ Pass ☒ Fail ☐ N/A	Verify dampers are open. Linears not supplying cold air. May be blockage Checked by: Theo Lagonikos on 02/22/2023 05:21 PM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:23 PM EST
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/22/2023 05:18 PM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:23 PM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/22/2023 05:18 PM EST
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	97F Checked by: Theo Lagonikos on 02/23/2023 10:16 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 10:56 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 10:56 AM EST

Spirit Airlines
IAH - Warehouse and Hanger

(E) VAV-15 (RCx VAV - Variable Air Volume Terminal...
FPT-16

Related Issues

#: 23 Asset: (E) VAV-15 Code: FPT-16-001 Status: Open Priority: Low

Reported By: Theo Lagonikos on 02/22/2023

Assigned to : Unassigned

Description:

Verify dampers are open. Linears not supplying cold air. May be blockage

Response(s):

None

END TEST

FPT-17: (E) VAV-16 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-16
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>ACCOUNTING 216



Set Points		Setpoint		Alarm	
Parameter	Description / Trend Information	Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:57 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:57 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:57 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:26 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 05:27 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 05:28 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:28 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 05:28 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 05:28 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 05:36 PM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 05:27 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 05:36 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 05:27 PM EST			
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	96F
Checked by: Theo Lagonikos on 02/23/2023 10:17 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:57 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:57 AM EST			

END TEST

FPT-18: (E) VAV-17 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-17
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>COFFEE 219



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					



Spirit Airlines
IAH - Warehouse and Hanger

(E) VAV-17 (RCx VAV - Variable Air Volume Terminal...
FPT-18

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- 1 Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
- 2 VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
- 3 HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve
Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
- 4 Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
- 5 Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
- 6 Heating temperature setpoint ☒ Pass ☐ Fail ☐ N/A Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint
Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
- 7 Min Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
- 8 Max Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 10:58 AM EST

#	Question	Answer	Details
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:58 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:52 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 05:37 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 05:38 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:38 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 05:38 PM EST
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:38 PM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:41 PM EST
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#	Question	Answer	Details
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/22/2023 05:41 PM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:41 PM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 05:41 PM EST
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	95F Checked by: Theo Lagonikos on 02/23/2023 10:17 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 10:58 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 10:58 AM EST

END TEST

FPT-19: (E) VAV-18 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-18
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>CONF RM 218



Set Points		Setpoint		Alarm	
Parameter	Description / Trend Information	Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 10:59 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 10:59 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 10:59 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:43 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 05:43 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 05:43 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:43 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 05:43 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 05:43 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 05:51 PM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 05:51 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 05:51 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 05:51 PM EST			
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	95F
Checked by: Theo Lagonikos on 02/23/2023 10:18 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 10:59 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 10:59 AM EST			

END TEST

FPT-20: (E) VAV-19 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-19
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>TRAINING DEVELOPMENT 224



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve
Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
- Heating temperature setpoint ☒ Pass ☐ Fail ☐ N/A Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint
Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
- Min Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
- Max Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 11:00 AM EST

#	Question	Answer	Details
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 11:00 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:16 AM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/23/2023 08:16 AM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/23/2023 08:16 AM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:16 AM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/23/2023 08:16 AM EST
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	57 deg F Checked by: Zach Yohon on 03/06/2023 11:06 AM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 09:36 AM EST
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#	Question	Answer	Details
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/23/2023 08:25 AM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:25 AM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow Checked by: Theo Lagonikos on 02/23/2023 08:25 AM EST
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	97F Checked by: Theo Lagonikos on 02/23/2023 09:34 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 11:00 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 11:00 AM EST

END TEST

FPT-21: (E) VAV-20 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-20
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>PILOTS OFFICE



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve
Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
- Heating temperature setpoint ☒ Pass ☐ Fail ☐ N/A Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint
Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
- Min Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
- Max Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 11:04 AM EST

#	Question	Answer	Details
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 11:04 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:31 AM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/23/2023 08:31 AM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/23/2023 08:32 AM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:32 AM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/23/2023 08:32 AM EST
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:32 AM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:44 AM EST
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#	Question	Answer	Details
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/23/2023 08:33 AM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:34 AM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow Checked by: Theo Lagonikos on 02/23/2023 08:33 AM EST
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	97F Checked by: Theo Lagonikos on 02/23/2023 09:26 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 11:04 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 11:04 AM EST

END TEST

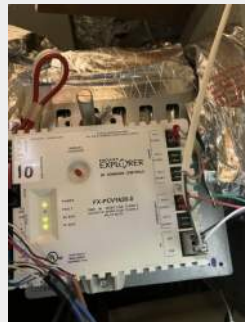
FPT-22: (E) VAV-21 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-21
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>OFFICE 230



Set Points		Setpoint		Alarm	
Parameter	Description / Trend Information	Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire			
#	Question	Answer	Details
Graphics Monitoring Points			
1	Room Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
2	VAV Flow (CFM)	☐ Pass ☐ Fail ☒ N/A	No CFM reading. Controller does have option for airflow measuring Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
3	HW Reheat Valve Position	☐ Pass ☐ Fail ☒ N/A	No valve position feedback. Only bms command to valve Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
4	Min CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
5	Max CFM	☐ Pass ☒ Fail ☐ N/A	No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor Checked by: Zach Yohon on 03/06/2023 11:07 AM EST

#	Question	Answer	Details
6	Heating temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
7	Min Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
8	Max Heating CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:07 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 11:07 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 06:01 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/22/2023 05:59 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/22/2023 05:59 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 05:59 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/22/2023 05:59 PM EST

#	Question	Answer	Details
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 05:59 PM EST			
Demonstrate call for heating: Simulate space temperature lower than setpoint			
1	VAV fan energizes	☐ Pass ☒ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 06:01 PM EST			
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode.
Checked by: Theo Lagonikos on 02/22/2023 06:00 PM EST			
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 06:00 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow
Checked by: Theo Lagonikos on 02/22/2023 06:00 PM EST			
5	Record discharge air temperature	☒ Pass ☐ Fail ☐ N/A	103F
Checked by: Theo Lagonikos on 02/23/2023 10:18 AM EST			
Alarms and Safeties			
1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed
Checked by: Zach Yohon on 03/06/2023 11:07 AM EST			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested
Checked by: Zach Yohon on 03/06/2023 11:07 AM EST			

END TEST

FPT-23: (E) VAV-22 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-22
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>IT RM 231



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No valve position feedback. Only bms command to valve
Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
- Heating temperature setpoint ☒ Pass ☐ Fail ☐ N/A Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint
Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
- Min Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
- Max Heating CFM ☐ Pass ☒ Fail ☐ N/A
Checked by: Zach Yohon on 03/06/2023 11:08 AM EST

#	Question	Answer	Details
9	Cooling temperature setpoint	☒ Pass ☐ Fail ☐ N/A	Currently programmed 72F with single setpoint and +/- 3 deg F local control. Recommend reprogramming to dual setpoint Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/06/2023 11:08 AM EST

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:49 AM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	Fan disables during cooling Checked by: Theo Lagonikos on 02/23/2023 08:51 AM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Damper modulates but no min/max setpoint programmed Checked by: Theo Lagonikos on 02/23/2023 08:52 AM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:52 AM EST
4	Zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	Box cannot measure airflow Checked by: Theo Lagonikos on 02/23/2023 08:51 AM EST
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:52 AM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 08:55 AM EST
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Spirit Airlines
IAH - Warehouse and Hanger

(E) VAV-22 (RCx VAV - Variable Air Volume Terminal...
FPT-23

#	Question	Answer	Details
2	Damper modulates to minimum position	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Checked by: Theo Lagonikos on 02/23/2023 08:51 AM EST
3	Hot water reheat valve modulates to maintain space temperature	☒ Pass ☐ Fail ☐ N/A	 Checked by: Theo Lagonikos on 02/23/2023 08:55 AM EST
4	Minimum zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Damper goes to full closed. No min position set. No OA during heating mode. Box cannot measure airflow Checked by: Theo Lagonikos on 02/23/2023 08:51 AM EST
5	Record discharge air temperature	☐ Pass ☒ Fail ☐ N/A	68F Checked by: Theo Lagonikos on 02/23/2023 09:29 AM EST

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No high/low zone temp alarms currently programmed Checked by: Zach Yohon on 03/06/2023 11:08 AM EST
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	Currently no fan status. Can be added if CT installed. Then can add fan mismatch alarm if requested Checked by: Zach Yohon on 03/06/2023 11:08 AM EST

Related Issues

#: 25	Asset: (E) VAV-22	Code: FPT-23-001	Status: Open	Priority: Low
Reported By: Theo Lagonikos on 02/23/2023				
Assigned to : Unassigned				
Description: Supply Air Temp is not rising during heating				
Response(s): None				

END TEST

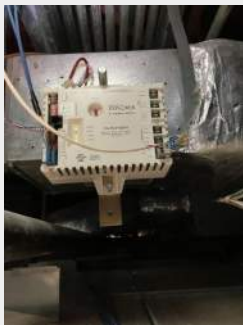
FPT-24: (E) VAV-23 (RCx VAV - Variable Air Volume Terminal Unit)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) VAV-23
Type	VARIABLE AIR VOLUME BOX
Description	VARIABLE AIR VOLUME BOX
Discipline	HVAC
Location(s)	HANGAR>SHOP 128



Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Min CFM Setpoint					
Max CFM Setpoint					
Occupied Cooling Setpoint					
Unoccupied Cooling Setpoint					

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Questionnaire

#	Question	Answer	Details
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Graphics Monitoring Points

- Room Temperature ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
- VAV Flow (CFM) ☐ Pass ☐ Fail ☒ N/A No CFM reading. Controller does have option for airflow measuring
Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
- HW Reheat Valve Position ☐ Pass ☐ Fail ☒ N/A No heating valve installed with unit.
Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
- Min CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
- Max CFM ☐ Pass ☒ Fail ☐ N/A No AFM device. Box not programmed for min/max airflow setting. Balancer to provide settings to controls contractor
Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
- Heating temperature setpoint ☐ Pass ☐ Fail ☒ N/A
Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
- Min Heating CFM ☐ Pass ☐ Fail ☒ N/A
Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
- Max Heating CFM ☐ Pass ☐ Fail ☒ N/A
Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
- Cooling temperature setpoint ☒ Pass ☐ Fail ☐ N/A
Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT

#	Question	Answer	Details
10	Min Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
11	Max Cooling CFM	☐ Pass ☒ Fail ☐ N/A	Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT
12	Discharge air temperature	☐ Pass ☒ Fail ☐ N/A	No DAT sensor Checked by: Zach Yohon on 03/20/2023 11:59 AM EDT

Occupied Mode

1	Enable AHU-1	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 03:40 PM EST
Demonstrate call for cooling: Simulate space temperature higher than setpoint.			
1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	No fan Checked by: Theo Lagonikos on 02/23/2023 03:40 PM EST
2	Damper modulates between minimum and maximum airflow to maintain CFM	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 03:45 PM EST
3	Zone temperature setpoint maintained	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 03:45 PM EST
4	Zone CFM setpoint maintained	☐ Pass ☒ Fail ☐ N/A	Lee Thompson to confirm if vav has AFMS. Show in BMS Checked by: Theo Lagonikos on 02/23/2023 03:57 PM EST
5	Record discharge air temperature with temperature gun	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/23/2023 03:49 PM EST

Demonstrate call for heating: Simulate space temperature lower than setpoint

1	VAV fan energizes	☐ Pass ☐ Fail ☒ N/A	No heating Checked by: Theo Lagonikos on 02/23/2023 03:41 PM EST
2	Damper modulates to minimum position	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/23/2023 03:40 PM EST



Spirit Airlines
IAH - Warehouse and Hanger

(E) VAV-23 (RCx VAV - Variable Air Volume Terminal...
FPT-24

#	Question	Answer	Details
3	Hot water reheat valve modulates to maintain space temperature	☐ Pass ☐ Fail ☒ N/A	
Checked by: Theo Lagonikos on 02/23/2023 03:40 PM EST			
4	Minimum zone CFM setpoint maintained	☐ Pass ☐ Fail ☒ N/A	
Checked by: Theo Lagonikos on 02/23/2023 03:40 PM EST			
5	Record discharge air temperature	☐ Pass ☐ Fail ☒ N/A	
Checked by: Theo Lagonikos on 02/23/2023 03:40 PM EST			

Alarms and Safeties

1	Verify Alarm is sent to BMS if space temperature is not within 5F of space temperature setpoint for more than 10 minutes.	☐ Pass ☒ Fail ☐ N/A	
Checked by: Zach Yohon on 03/20/2023 12:00 PM EDT			
2	Verify alarm is sent to BMS if fan command does not match fan status.	☐ Pass ☒ Fail ☐ N/A	
Checked by: Zach Yohon on 03/20/2023 12:00 PM EDT			

END TEST

FPT-25: (E) AC-1 (RCx - DX Split)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) AC-1
Type	INDOOR AIR CONDITIONING UNIT
Description	INDOOR AIR CONDITIONING UNIT
Discipline	HVAC
Location(s)	WAREHOUSE>1ST FLOOR MECH RM



Sequence Of Operation

Spirit Airlines
IAH - Warehouse and Hanger

(E) AC-1 (RCx - DX Split)
FPT-25

THE SUPPLY FAN SHALL CYCLE ON/OFF WITH THE COMPRESSOR TO MAINTAIN A SPACE TEMPERATURE OF 72° F (ADJ.) UNDER ITS OWN CONTROL

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout *Factory Controls			

Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Cooling Setpoint (72° F)* ☒ Pass ☐ Fail ☐ N/A Set to 69



1676993691173.jpg

Checked by: Theo Lagonikos on 02/21/2023 10:32 AM EST

Devices and Outputs -
Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off, Open/Closed, 0%-100%, VFD Speed, etc.)

#	Question	Answer	Details
1	Enable/Disable	☒ Pass ☐ Fail ☐ N/A	



1676994514441.jpg



1676994534121.jpg

Checked by: Theo Lagonikos on 02/21/2023 10:44 AM EST

Ductless Split-System Functional Performance Testing

Off Mode

1	Disable system via local controls and verify: 1. Supply Fan is off. 2. Associated Compressor is off	☒ Pass ☐ Fail ☐ N/A	
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Checked by: Theo Lagonikos on 02/21/2023 10:37 AM EST

Cooling Mode

1	Decrease the space temperature cooling setpoint (72°F) below space temperature and verify: 1. Unit shall enable in Cooling Mode. 2. Supply fan starts. 3. Internal controls within the unit shall cycle supply fan to maintain space air temperature setpoint.	☒ Pass ☐ Fail ☐ N/A	53 deg F SAT. Deflectors has trouble opening/closing
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Checked by: Theo Lagonikos on 02/21/2023 10:50 AM EST

Condensate Pump Operation

#	Question	Answer	Details
1	Manually modulate condensate pump float level sensor at the unit and verify: 1. Condensate pump enables and is pumping water out of the pan 2. Condensate pump disables once the water level has lowered below the alarm setting 3. If unit is connected to BMS, verify high condensate alarm is sent. 4. Unit restarts after condensate overflow condition has been rectified.	☐ Pass ☐ Fail ☒ N/A	Gravity drained. Could not test internal overflow switch if it exists



1676994747865.jpg

Checked by: Theo Lagonikos on 02/21/2023 10:51 AM EST

END TEST

Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-13 (RCx - Ventilation Exhaust)
FPT-26

FPT-26: (E) EF-13 (RCx - Ventilation Exhaust)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) EF-13
Type	EXHAUST FAN
Description	EXHAUST FAN
Discipline	HVAC
Location(s)	HANGAR>PUMP RM 100B

System Description

Ventilation exhaust fan controlled by wall mounted thermostat

Questionnaire

#	Question	Answer	Details
Sequence of Operation			

#	Question	Answer	Details
1	Prior to functional testing, record current temperature setpoint.	☒ Pass ☐ Fail ☐ N/A	60 deg f



1676998848177.jpg



1676998862227.jpg



1676998904908.jpg

Checked by: Theo Lagonikos on 02/21/2023 11:56 AM EST

Enable exhaust fan - toggle exhaust fan thermostat dial to max

1	Exhaust fan shall turn on	☒ Yes ☐ No ☐ N/A	
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Checked by: Theo Lagonikos on 02/21/2023 11:56 AM EST

2	Associated motorized OA intake damper opens	☐ Pass ☒ Fail ☐ N/A	Damper remains open at all times.
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Checked by: Theo Lagonikos on 03/06/2023 09:43 AM EST

Disable exhaust fan - toggle exhaust fan thermostat dial to min

1	Exhaust fan shall turn off	☒ Yes ☐ No ☐ N/A	
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Checked by: Theo Lagonikos on 02/21/2023 11:56 AM EST

2	Associated motorized OA intake damper closes	☐ Pass ☒ Fail ☐ N/A	Actuator was not able to close damper.
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Checked by: Theo Lagonikos on 03/06/2023 09:43 AM EST

Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-13 (RCx - Ventilation Exhaust)
FPT-26

Related Issues

#: 6 Asset: (E) EF-13 Code: FPT-26-001 Status: Open Priority: Low

Reported By: Theo Lagonikos on 02/21/2023

Assigned to : Unassigned

Description:
Motorized damper does not close when EF is disabled

Response(s):
None

END TEST

Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-14 (RCx - Toilet Exhaust)
FPT-27

FPT-27: (E) EF-14 (RCx - Toilet Exhaust)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

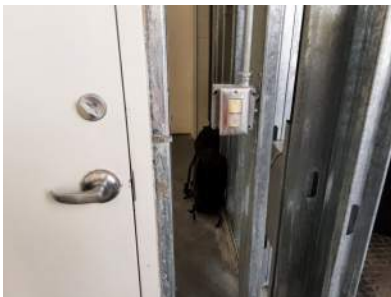
Unit #	(E) EF-14
Type	EXHAUST FAN
Description	EXHAUST FAN
Discipline	HVAC
Location(s)	HANGAR>BATHROOM 100A

System Description

Toilet exhaust fan controlled by wall mounted toggle switch

Questionnaire

#	Question	Answer	Details
Enable exhaust fan - toggle switch to "On" position			
1	Exhaust fan shall turn on	☒ Yes ☐ No ☐ N/A	EF controlled by light switch / occupancy sensor.



1676998162312.jpg



1676998186912.jpg

Checked by: Theo Lagonikos on 02/21/2023 11:49 AM EST

Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-14 (RCx - Toilet Exhaust)
FPT-27

#	Question	Answer	Details
Disable exhaust fan - toggle switch to "Off" position			
1	Exhaust fan shall turn off	☒ Yes ☐ No ☐ N/A	

Checked by: Theo Lagonikos on 02/21/2023 11:47 AM EST

END TEST

Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-8 (RCx - Exhaust Fan)
FPT-58

FPT-58: (E) EF-8 (RCx - Exhaust Fan)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) EF-8
Type	EXHAUST FAN
Description	EXHAUST FAN (Roof Mounted)
Discipline	HVAC
Location(s)	HANGAR>ADMIN ROOF



System Description

Roof exhaust fan

Sequence Of Operation

Runs continuous 24/7



Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-8 (RCx - Exhaust Fan)
FPT-58

Questionnaire

#	Question	Answer	Details
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Verify exhaust fan is enabled and operational.

1 Exhaust fan is operational ☒ Yes ☐ No ☐ N/A

Checked by: Theo Lagonikos on 03/03/2023 02:57 PM EST

Exhaust fan is visible in BMS and can be enabled/disabled in BMS

1 Exhaust fan can be enabled/disabled in BMS ☐ Yes ☐ No ☒ N/A

Checked by: Theo Lagonikos on 03/03/2023 02:57 PM EST

END TEST

Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-9 (RCx - Exhaust Fan)
FPT-59

FPT-59: (E) EF-9 (RCx - Exhaust Fan)

Test Type: Functional Performance Testing

Status: Draft

Asset Summary

Unit #	(E) EF-9
Type	EXHAUST FAN
Description	EXHAUST FAN
Discipline	HVAC
Location(s)	HANGAR>ADMIN ROOF



System Description

Roof exhaust fan

Sequence Of Operation

Runs continuous 24/7



Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-9 (RCx - Exhaust Fan)
FPT-59

Questionnaire

#	Question	Answer	Details
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Verify exhaust fan is enabled and operational.

1 Exhaust fan is operational ☒ Yes ☐ No ☐ N/A

Checked by: Theo Lagonikos on 03/03/2023 03:06 PM EST

Exhaust fan is visible in BMS and can be enabled/disabled in BMS

1 Exhaust fan can be enabled/disabled in BMS ☐ Yes ☐ No ☒ N/A

Checked by: Theo Lagonikos on 03/03/2023 03:06 PM EST

END TEST

Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-2-W (RCx - Toilet Exhaust)
FPT-28

FPT-28: (E) EF-2-W (RCx - Toilet Exhaust)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) EF-2-W
Type	EXHAUST FAN
Description	EXHAUST FAN
Discipline	HVAC
Location(s)	WAREHOUSE>2ND FLR STORAGE RM



System Description

Toilet exhaust fan controlled by wall mounted occupancy sensor.

Questionnaire

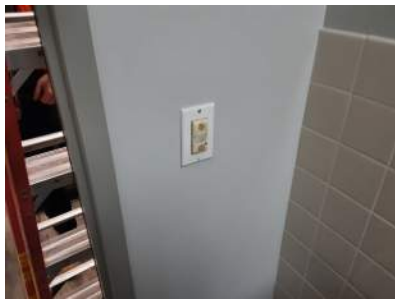
#	Question	Answer	Details
Sequence of Operation			

1 Verify if fan is integrated with BMS. ☐ Pass ☐ Fail ☒ N/A Not on BMS

Checked by: Theo Lagonikos on 02/21/2023 11:06 AM EST

Enable exhaust fan - toggle switch to "On" position, if controlled by occupancy sensor, vacate room for 15 minutes

1 Exhaust fan shall turn on ☒ Yes ☐ No ☐ N/A



1676995630898.jpg



1676995619707.jpg

Checked by: Theo Lagonikos on 02/21/2023 11:06 AM EST

Disable exhaust fan - toggle switch to "Off" position

1 Exhaust fan shall turn off ☒ Yes ☐ No ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 11:09 AM EST

END TEST

FPT-29: (E) GRH (RCx - GRH Zone 1)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) GRH
Type	GAS RADIANT HEATER
Description	GAS RADIANT HEATER
Discipline	HVAC
Location(s)	HANGAR>AREAS



Sequence Of Operation

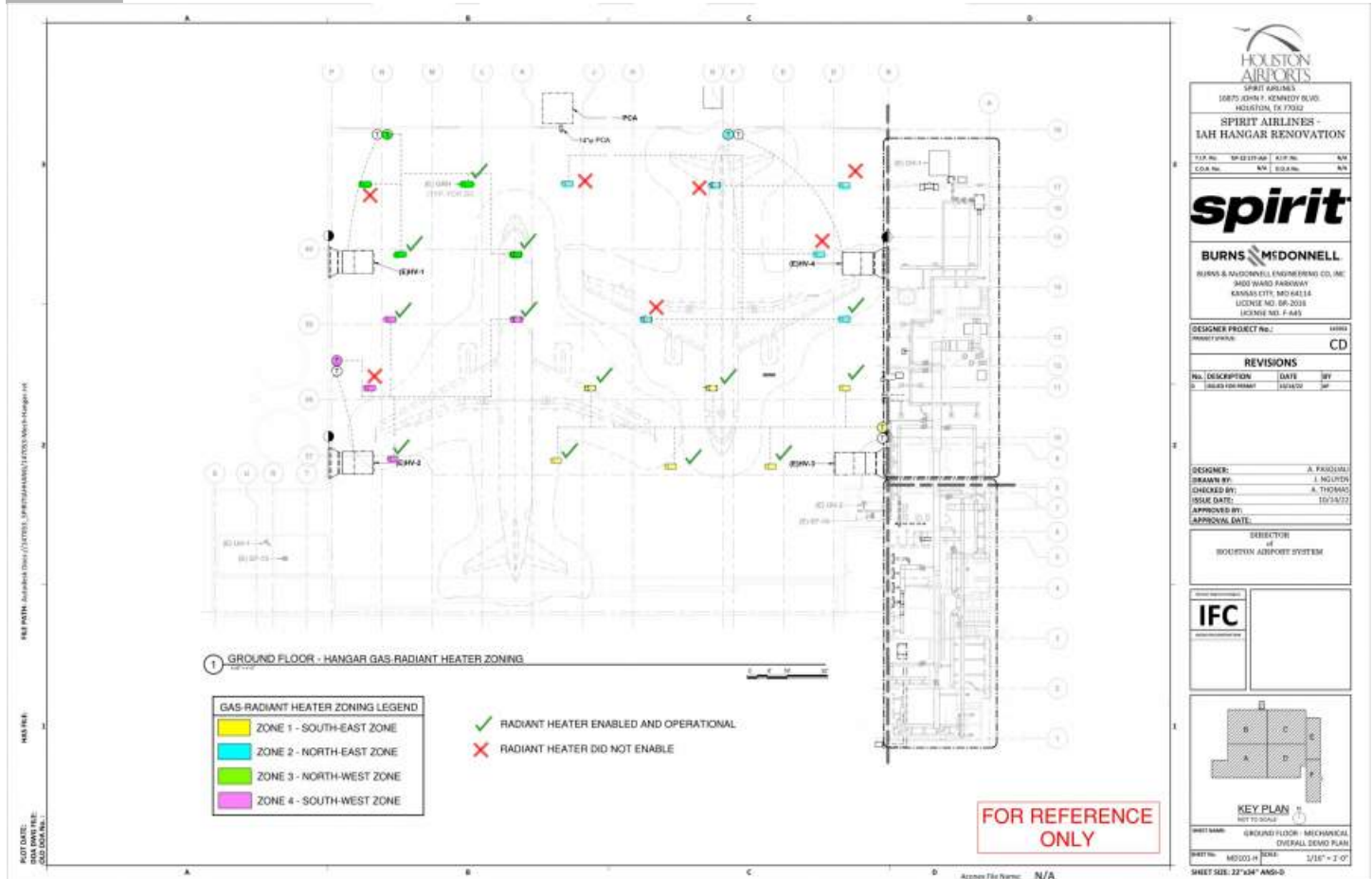
INFRARED UNIT HEATER

1. THE HEATER SHALL STAGE HEAT UNDER ITS OWN CONTROLS TO MAINTAIN A SPACE TEMPERATURE OF 55° F TO 60° F (ADJ.).
2. UPON THE ACTIVATION OF THE FIRE ALARM RELAY, THE SYSTEM SHALL SHUTDOWN ALL BURNERS AND VACCUM PUMPS.
3. THE INFARED HEATERS SHALL BE INTERLOCKED WITH THE FOAM SUPPRESSION SYSTEM, AND SHUT DOWN UPON ACTIVATION OF THE FOAM SYSTEM

**Spirit Airlines
IAH - Warehouse and Hanger**

(E) GRH (RCx - GRH Zone 1)
FPT-29

Diagrams



Spirit Airlines
IAH - Warehouse and Hanger

(E) GRH (RCx - GRH Zone 1)
FPT-29

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

1 Verify if heater has multiple stages. ☐ Pass ☐ Fail ☒ N/A

Checked by: Theo Lagonikos on 02/21/2023 12:36 PM EST

Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Heating Setpoint (55/60°F) (Local Control) ☒ Pass ☐ Fail ☐ N/A SE zone



1677001030621.jpg

Checked by: Theo Lagonikos on 02/21/2023 12:38 PM EST

Sensors -
Confirm the installed location of each sensor is near the design location.

1 Wall Mounted Space Thermostat ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 12:36 PM EST

IRH Functional Performance Testing

Off Mode

1 Simulate the IRH in Off mode via the controller and verify:
1. All units are disabled.
2. Associated vacuum pumps are disabled ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 12:36 PM EST

General Operation

#	Question	Answer	Details
1	Simulate the space temperature below the space temperature setpoint of 55°F and verify: 1. Unit shall enable. Including vacuum pumps (if applicable) 2. Heater shall stage heat to maintain a space temperature between 55°F and 60°F. 3. Unit disables once space temperature is satisfied	☒ Pass ☐ Fail ☐ N/A	

Checked by: Theo Lagonikos on 02/21/2023 12:36 PM EST

END TEST

FPT-30: (E) GRH (RCx - GRH Zone 2)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) GRH
Type	GAS RADIANT HEATER
Description	GAS RADIANT HEATER
Discipline	HVAC
Location(s)	HANGAR>AREAS

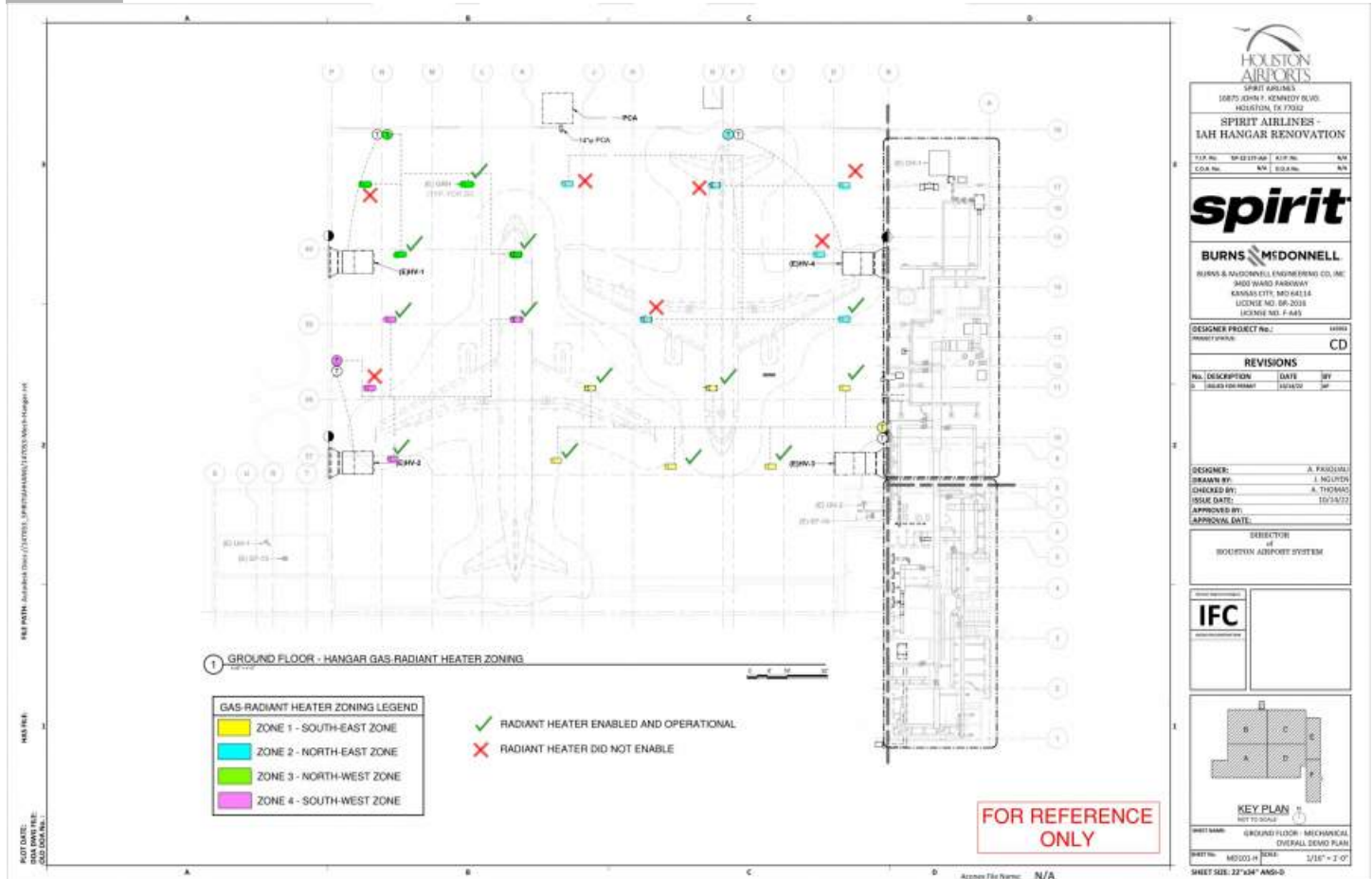


Sequence Of Operation

INFRARED UNIT HEATER

1. THE HEATER SHALL STAGE HEAT UNDER ITS OWN CONTROLS TO MAINTAIN A SPACE TEMPERATURE OF 55° F TO 60° F (ADJ.).
2. UPON THE ACTIVATION OF THE FIRE ALARM RELAY, THE SYSTEM SHALL SHUTDOWN ALL BURNERS AND VACCUM PUMPS.
3. THE INFARED HEATERS SHALL BE INTERLOCKED WITH THE FOAM SUPPRESSION SYSTEM, AND SHUT DOWN UPON ACTIVATION OF THE FOAM SYSTEM

Diagrams



Spirit Airlines
IAH - Warehouse and Hanger

(E) GRH (RCx - GRH Zone 2)
FPT-30

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

1 ☐ Pass ☐ Fail ☒ N/A

Checked by: Theo Lagonikos on 02/21/2023 12:51 PM EST

Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 ☒ Pass ☐ Fail ☐ N/A



1677001915548.jpg

Checked by: Theo Lagonikos on 02/21/2023 12:54 PM EST

Sensors -
Confirm the installed location of each sensor is near the design location.

1 ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 12:52 PM EST

IRH Functional Performance Testing

Off Mode

1 ☐ Pass ☒ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 12:52 PM EST

General Operation

#	Question	Answer	Details
1	Simulate the space temperature below the space temperature setpoint of 55°F and verify: 1. Unit shall enable. Including vacuum pumps (if applicable) 2. Heater shall stage heat to maintain a space temperature between 55°F and 60°F. 3. Unit disables once space temperature is satisfied	☐ Pass ☒ Fail ☐ N/A	Only 1 GRH was enabled



1677001992118.jpg

Checked by: Theo Lagonikos on 02/21/2023 12:53 PM EST

Related Issues

#: 7	Asset: (E) GRH	Code: FPT-30-001	Status: Open	Priority: Low
Reported By: Theo Lagonikos on 02/21/2023				
Assigned to : Unassigned				
Description: Only 1 GRHs enabled				
Response(s): None				

END TEST

FPT-31: (E) GRH (RCx - GRH Zone 3)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) GRH
Type	GAS RADIANT HEATER
Description	GAS RADIANT HEATER
Discipline	HVAC
Location(s)	HANGAR>AREAS

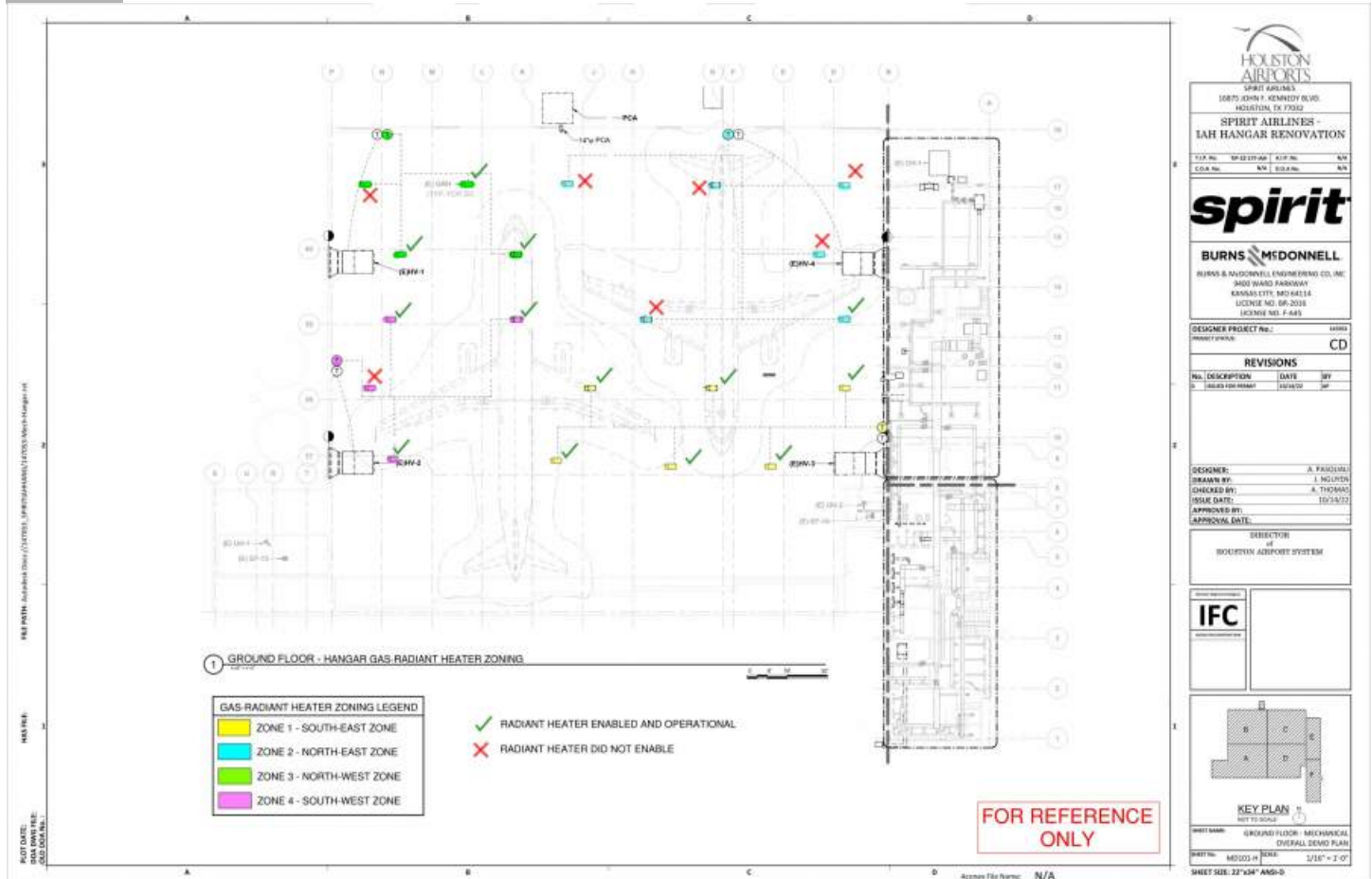


Sequence Of Operation

INFRARED UNIT HEATER

1. THE HEATER SHALL STAGE HEAT UNDER ITS OWN CONTROLS TO MAINTAIN A SPACE TEMPERATURE OF 55° F TO 60° F (ADJ.).
2. UPON THE ACTIVATION OF THE FIRE ALARM RELAY, THE SYSTEM SHALL SHUTDOWN ALL BURNERS AND VACCUM PUMPS.
3. THE INFARED HEATERS SHALL BE INTERLOCKED WITH THE FOAM SUPPRESSION SYSTEM, AND SHUT DOWN UPON ACTIVATION OF THE FOAM SYSTEMTHE FOAM SYSTEM

Diagrams



Spirit Airlines
IAH - Warehouse and Hanger

(E) GRH (RCx - GRH Zone 3)
FPT-31

Questionnaire

#	Question	Answer	Details
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Point-to-Point Checkout

1 Verify if heater has multiple stages. ☐ Pass ☐ Fail ☒ N/A

Checked by: Theo Lagonikos on 02/21/2023 01:01 PM EST

Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Heating Setpoint (55/60°F) (Local Control) ☒ Pass ☐ Fail ☐ N/A NW zone



1677002542272.jpg

Checked by: Theo Lagonikos on 02/21/2023 01:02 PM EST

Sensors -
Confirm the installed location of each sensor is near the design location.

1 Wall Mounted Space Thermostat ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 01:03 PM EST

IRH Functional Performance Testing

Off Mode

1 Simulate the IRH in Off mode via the controller and verify:
1. All units are disabled.
2. Associated vacuum pumps are disabled ☐ Pass ☒ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 01:03 PM EST

General Operation

#	Question	Answer	Details
1	Simulate the space temperature below the space temperature setpoint of 55°F and verify: 1. Unit shall enable. Including vacuum pumps (if applicable) 2. Heater shall stage heat to maintain a space temperature between 55°F and 60°F. 3. Unit disables once space temperature is satisfied	☐ Pass ☒ Fail ☐ N/A	1 of 4 GRHs did not enable

Checked by: Theo Lagonikos on 02/21/2023 01:03 PM EST

Related Issues

#: 8	Asset: (E) GRH	Code: FPT-31-001	Status: Open	Priority: Low
Reported By: Theo Lagonikos on 02/21/2023				
Assigned to : Unassigned				
Description: 1 of 4 GRHs did not enable				
Response(s): None				

END TEST

FPT-32: (E) GRH (RCx - GRH Zone 4)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) GRH
Type	GAS RADIANT HEATER
Description	GAS RADIANT HEATER
Discipline	HVAC
Location(s)	HANGAR>AREAS



Sequence Of Operation

INFRARED UNIT HEATER

1. THE HEATER SHALL STAGE HEAT UNDER ITS OWN CONTROLS TO MAINTAIN A SPACE TEMPERATURE OF 55° F TO 60° F (ADJ.).
2. UPON THE ACTIVATION OF THE FIRE ALARM RELAY, THE SYSTEM SHALL SHUTDOWN ALL BURNERS AND VACCUM PUMPS.
3. THE INFARED HEATERS SHALL BE INTERLOCKED WITH THE FOAM SUPPRESSION SYSTEM, AND SHUT DOWN UPON ACTIVATION OF THE FOAM SYSTEM

Spirit Airlines
IAH - Warehouse and Hanger

(E) GRH (RCx - GRH Zone 4)
FPT-32

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

1 Verify if heater has multiple stages. ☐ Pass ☐ Fail ☒ N/A

Checked by: Theo Lagonikos on 02/21/2023 01:07 PM EST

Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Heating Setpoint (55/60°F) (Local Control) ☒ Pass ☐ Fail ☐ N/A SW zone



1677002873486.jpg

Checked by: Theo Lagonikos on 02/21/2023 01:07 PM EST

Sensors -
Confirm the installed location of each sensor is near the design location.

1 Wall Mounted Space Thermostat ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 01:07 PM EST

IRH Functional Performance Testing

Off Mode

1 Simulate the IRH in Off mode via the controller and verify:
1. All units are disabled.
2. Associated vacuum pumps are disabled ☐ Pass ☒ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 01:14 PM EST

General Operation

Spirit Airlines
IAH - Warehouse and Hanger

(E) GRH (RCx - GRH Zone 4)
FPT-32

#	Question	Answer	Details
1	Simulate the space temperature below the space temperature setpoint of 55°F and verify: 1. Unit shall enable. Including vacuum pumps (if applicable) 2. Heater shall stage heat to maintain a space temperature between 55°F and 60°F. 3. Unit disables once space temperature is satisfied	☐ Pass ☒ Fail ☐ N/A	1 of 4 GRHs did not enable

Checked by: Theo Lagonikos on 02/21/2023 01:14 PM EST

Related Issues

#: 9	Asset: (E) GRH	Code: FPT-32-001	Status: Open	Priority: Low
Reported By: Theo Lagonikos on 02/21/2023				
Assigned to : Unassigned				
Description: 1 of 4 GRHs did not enable				
Response(s): None				

END TEST

FPT-33: Chilled Water System (CHW System)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

System Name	Chilled Water System
Discipline	HVAC
Location(s)	HANGAR
Related Equipment	(E) CH-1, (E) CHWP-2, (E) CHWP-1

Questionnaire

#	Question	Answer	Details
Sequence of Operation			

Schedule the chilled water system OFF through the BAS and verify the following:

1 CH-1 is off ☒ Pass ☐ Fail ☐ N/A

Checked by: Zach Yohon on 02/23/2023 12:55 PM EST

2 Associated chilled water pumps (CHWP 1 & 2) are off ☒ Pass ☐ Fail ☐ N/A When OA is above OA lockout setpoint. Chiller SOO is enabled. When RA temp is above RA lockout, chiller & chw pumps will run.

Checked by: Zach Yohon on 02/23/2023 12:55 PM EST

Normal Operation

1 Verify chiller is provided with a local controller that controls all operational and protective functions of the chiller. ☒ Pass ☐ Fail ☐ N/A

Checked by: Zach Yohon on 02/23/2023 12:55 PM EST

2 Verify the BAS will allow the operator to start the chiller, adjust water temperature set points, and start/sequence the chilled water pumps. ☒ Pass ☐ Fail ☐ N/A BMS can start/stop chiller. Water temperature adjustment is changed via local controller. BMS can start/stop CHW pumps.

Checked by: Zach Yohon on 02/23/2023 12:55 PM EST



Spirit Airlines
IAH - Warehouse and Hanger

Chilled Water System (CHW System)
FPT-33

#	Question	Answer	Details
3	Verify chilled water pumps are in "auto" mode.	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 12:55 PM EST
4	Verify Chilled Water Set Point.	☒ Pass ☐ Fail ☐ N/A	44 deg F, done at chiller controller Checked by: Zach Yohon on 02/23/2023 12:55 PM EST
Enable chiller controls so that the unit starts in "auto" mode. Observe initial startup and operation. Verify the following information.			
1	Chilled water pump starts	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 01:03 PM EST
2	Chiller controller determines if cooling is needed (evaporator leaving temperature is above the start differential set point)	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 01:03 PM EST
3	Chiller verifies required chilled water flow is present	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 01:03 PM EST
4	Lead compressor starts assuming no lock out conditions exist (for example restart inhibit)	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 01:12 PM EST
After the CHW Supply Temperature setpoint is met, decrease the load on the chiller so that it unloads down to a single compressor and record the following:			
1	Chilled Water Outlet Temp	☐ Pass ☒ Fail ☐ N/A	Chilled water supply thermometer not functioning Checked by: Zach Yohon on 02/23/2023 01:21 PM EST
2	Chiller Running Current	☐ Pass ☐ Fail ☒ N/A	Checked by: Zach Yohon on 02/23/2023 01:21 PM EST
Simulate a chiller shutdown signal through the control system and verify the following.			
1	The chiller should be disabled and begin its shutdown sequence.	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 01:03 PM EST
2	The compressor(s) should unload and the chiller should shutdown.	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 01:03 PM EST

Safeties & Alarms

Chiller Failure: Verify that the following conditions mark the chiller as "failed" and turns on a chiller failure indicator alarm.

#	Question	Answer	Details
1	Chiller status cannot be confirmed as "on" during start sequence	☐ Pass ☒ Fail ☐ N/A	No alarm in BMS programming Checked by: Zach Yohon on 02/23/2023 01:03 PM EST
2	Chilled water flow is lost after startup	☐ Pass ☐ Fail ☒ N/A	Alarm done at local controller. Did not test when on-stie Checked by: Zach Yohon on 03/20/2023 12:07 PM EDT
3	Chiller status cannot be confirmed as "off" during shutdown	☐ Pass ☒ Fail ☐ N/A	No alarm in BMS programming Checked by: Zach Yohon on 02/23/2023 01:03 PM EST
4	After the failure sequence is verified, ensure that the chillers can be reset via a reset push button on both the chiller and system control panels. Also, verify that a "system reset" pushbutton is available through the plant controls to reset the operation of the failed chiller.	☐ Pass ☒ Fail ☐ N/A	Chiller cannot be manually reset. Buttons on chiller local controller display are not functioning. Checked by: Zach Yohon on 02/23/2023 01:03 PM EST

Trending

1	Trending and graphic display of trends are programmable and display on the chiller control panel screen. Attach Screenshot sample	☐ Pass ☐ Fail ☒ N/A	No trending Checked by: Zach Yohon on 02/23/2023 01:03 PM EST
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Points List

Analog I / O

1	Actual Capacity (%RLA)	☐ Pass ☐ Fail ☒ N/A	 Checked by: Zach Yohon on 02/23/2023 01:11 PM EST
2	Active CHW Setpoint	☐ Pass ☐ Fail ☒ N/A	At local controller Checked by: Zach Yohon on 02/23/2023 01:11 PM EST
3	Active Current Limit Setpoint	☐ Pass ☐ Fail ☒ N/A	 Checked by: Zach Yohon on 02/23/2023 01:11 PM EST

Digital I / O

1	Chiller Enable/Disable	☒ Pass ☐ Fail ☐ N/A	 Checked by: Zach Yohon on 02/23/2023 01:11 PM EST
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Spirit Airlines
IAH - Warehouse and Hanger

Chilled Water System (CHW System)
FPT-33

#	Question	Answer	Details
2	Chiller Run Request	☐ Pass ☐ Fail ☒ N/A	Checked by: Zach Yohon on 02/23/2023 01:11 PM EST
3	CHWP-1 Enable	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 01:11 PM EST
4	CHWP-2 Enable	☒ Pass ☐ Fail ☐ N/A	Checked by: Zach Yohon on 02/23/2023 01:11 PM EST

Alarms

1	Chiller Failure	☐ Pass ☐ Fail ☒ N/A	at local controller Checked by: Zach Yohon on 02/23/2023 01:11 PM EST
2	Manual Reset Alarm	☐ Pass ☐ Fail ☒ N/A	Checked by: Zach Yohon on 02/23/2023 01:11 PM EST
3	Motor Protection Status	☐ Pass ☐ Fail ☒ N/A	at local controller Checked by: Zach Yohon on 02/23/2023 01:11 PM EST
4	Auto Reset Alarm	☐ Pass ☐ Fail ☒ N/A	Checked by: Zach Yohon on 02/23/2023 01:11 PM EST
5	Informational Warning	☐ Pass ☐ Fail ☒ N/A	at local controller Checked by: Zach Yohon on 02/23/2023 01:11 PM EST

END TEST

FPT-34: Hot Water System (HHW System)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

System Name	Hot Water System
Discipline	HVAC
Location(s)	HANGAR
Related Equipment	(E) B-1, (E) HWP-1

Sequence Of Operation

HEATING WATER SYSTEM:

IF THE OUTSIDE AIR TEMPERATURE FALL BELOW SETPOINT (INITIALLY 70°F, 8°F DEADBAND, OFF AT 80°F):

1. ENABLE BOILER CONTROLLER TO ENABLE THE BOILER (B LR-1 OR BLR-2) AND THE PRIMARY PUMP (BCP-1 OR BCP-2) FOR THE LEAD BOILER.
2. WHEN A BOILER IS TURNED OFF, THE ASSOCIATED BOILER PUMP CONTINUES TO RUN FOR FIVE (5) MINUTES.
3. MODULATE BOILERS TO MAINTAIN HEATING WATER SUPPLY SETPOINT TEMPERATURE. SEE SCHEMATIC FOR PRELIMINARY RESET SCHEDULE.

BOILER MODULATION:

1. THE MODULATING BOILER SEQUENCER (INTEGRAL TO EACH BOILER) STAGES AND MODULATES THE OUTPUT FROM EACH BOILER. BOILER INTEGRAL CONTROLS OPERATE BOILER TO MAINTAIN HEATING WATER SUPPLY TEMPERATURE SETPOINT. BOILER MANUFACTURER TO PROVIDE A BACNET INTERFACE TO CONNECT TO THE BMS AND ALLOW FOR MONITORING OF STATUS, FIRING RATE, LEAVING WATER TEMPERATURE SETPOINT AND SUMMARY ALARM FOR EACH BOILER AND DISPLAY ON THE SYSTEM GRAPHICS PAGE.

PRIMARY HEATING WATER PUMPS (BCP-1 & BCP-2):

1. THE BOILER PUMPS SHALL OPERATE IN CONJUNCTION WITH THEIR ASSOCIATED BOILER.
2. START PUMP UPON FIRING SIGNAL FROM A BOILER. SET A 30 SECOND FIRING DELAY ON THE BOILER IGNITION FROM TIME OF FIRING SIGNAL TO BURNER IGNITION.
3. WHEN BOILER IS TURNED OFF AFTER OPERATION, THE ASSOCIATED BOILER PUMP SHALL REMAIN RUNNING FOR 5 MINUTES TO REMOVE RESIDUAL HEAT FROM BOILER.

PRIMARY HEATING WATER PUMPS FAILURE (BCP-1 & BCP-2):

IF A PRIMARY HEATING WATER PUMP FAILS:

1. SHUT DOWN THE ASSOCIATED BOILER.

2. ALARM AT THE BUILDING AUTOMATION DDC SYSTEM.
3. START THE LAG BOILER TO OPERATE AND ASSOCIATED PUMP.
4. WHEN THE FAILED PUMP/BOILER RE-START, RETURN TO THE ORIGINAL (BEFORE FAILURE) OPERATING SEQUENCE.

SECONDARY HEATING WATER PUMPS (HWCP-1 & HWCP-2):

1. THE SECONDARY HEATING WATER PUMPS OPERATE WHENEVER THERE IS A CALL FOR HEATING FROM THE BUILDING. INTEGRAL PUMP CONTROLS MODULATE PUMPS TO MAINTAIN DIFFERENTIAL PRESSURE SETPOINT IN THE PIPING SYSTEM (SETPOINT TO BE DETERMINED DURING SYSTEM BALANCING).

SECONDARY HEATING WATER PUMPS FAILURE (HWCP-1 & HWCP-2):

IF A SECONDARY HEATING WATER PUMP FAILS:

1. ALARM AT THE BUILDING AUTOMATION DDC SYSTEM.
2. START THE LAG PUMP TO OPERATE.
3. WHEN THE FAILED PUMP RE-START, RETURN TO THE ORIGINAL (BEFORE FAILURE) OPERATING SEQUENCE.

LEAD /LAG SELECTION:

LEAD/LAG SELECTION OF BOILERS AND PUMPS SHALL BE SCHEDULED TO CHANGE AUTOMATICALLY EACH WEEK (CHANGEOVER SCHEDULED AT 10:00AM EACH WEDNESDAY). LAG BOILER IS INTENDED TO OPERATE IN STAND-BY MODE, OR AS NEEDED.

SAFETIES:

ALARM AT THE BUILDING AUTOMATION DDC SYSTEM IF ANY OF THE FOLLOWING CONDITIONS EXIST:

1. INDICATE ALARM CONDITION UPON FAILURE OF BOILER BURNER AND/OR WHENEVER THE HWS TEMPERATURE FALLS BELOW THE ALARM SETPOINT (90°F, ADJUSTABLE) FOR MORE THAN 30 MINUTES WHILE THE BOILER IS RUNNING.
2. MONITOR THE SUMMARY ALARM CONTACT ON EACH BOILER, AND GENERATE AN ALARM UPON BOILER FAILURE.
3. MONITOR EACH PUMP WITH AN ADJUSTABLE ANALOG CURRENT SENSOR. GENERATE AN ALARM UPON PUMP FAILURE. PROVIDE A DIGITAL BACNET INTERFACE BETWEEN THE BAS AND THE BOILER SEQUENCER, TO ALLOW FOR MONITORING OF THE FOLLOWING ON ALL BOILERS: STATUS, FIRING RATE, LEAVING WATER TEMPERATURE SETPOINT, FLUE GAS TEMPERATURE AND SUMMARY ALARM.

Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Boiler 1 Supply Water Temperature Setpoint	°F				
Boiler 2 Supply Water Temperature Setpoint	°F				
Heating Water Supply Temperature Setpoint	°F				
Heating Water System Differential Pressure Setpoint	PSI				

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Heating Water Minimum Flow Setpoint	GPM				

Questionnaire

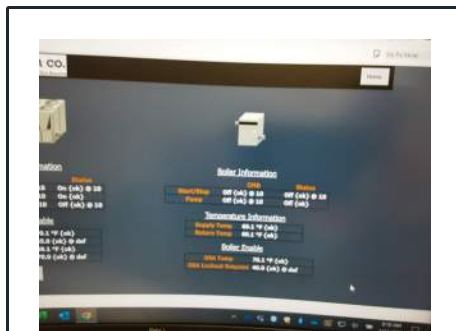
#	Question	Answer	Details
System Functional Performance Test (FPT): Heating Water System (HW)			

Reminder: Commissioning Agents are not to 'touch' or adjust settings on the equipment being tested. All adjustments to the equipment itself, or the BAS equipment controls, are to be performed by the installing contractor or the controls contractor. During testing activities ensure that adjustments to equipment and systems settings are done in coordination with all other disciplines associated with the equipment and with all individuals who may be working near the equipment or its associated system components.

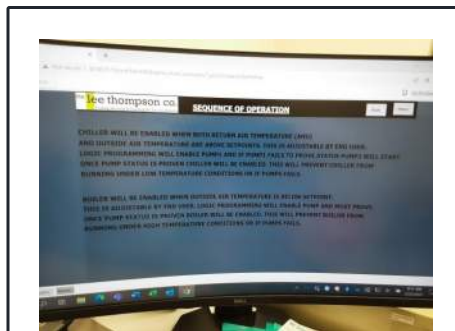
INITIAL CHECKS

Confirm all of the following points are visible in the BAS:

- 1 Boiler 1 Leaving Water Temperature ☒ Pass ☐ Fail ☐ N/A Lee Thompson to rename HWS. Looking at HWR



1677078662150.jpg



1677078711876.jpg

Checked by: Theo Lagonikos on 02/22/2023 10:24 AM EST

- 2 Heating Water Supply Setpoint ☒ Pass ☐ Fail ☐ N/A 160 F via setpoint integral in boiler

Checked by: Theo Lagonikos on 02/22/2023 10:04 AM EST

- 3 Boiler Entering Water Temperature ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/22/2023 10:06 AM EST

#	Question	Answer	Details
4	Outside Air Dry Bulb Temperature	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:05 AM EST
5	B- Firing Rate	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:06 AM EST
6	B- 1 Enable	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:06 AM EST
7	B-1 Status	☐ Pass ☒ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:29 AM EST
8	B-1 Alarm	☐ Pass ☐ Fail ☒ N/A	Only loss of communication alarm. Will need to add current transducer to get boiler status Checked by: Theo Lagonikos on 02/22/2023 10:28 AM EST
9	Heating Water Pump HWP-1 Enable	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:07 AM EST
10	Heating Water Pump HWP-1 Status	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:07 AM EST
11	Heating Water System Load	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:07 AM EST
Confirm 15 minute trends are collecting for all of the following points:			
1	Boiler 1 Enable	☐ Pass ☐ Fail ☒ N/A	No trending Checked by: Theo Lagonikos on 02/22/2023 10:08 AM EST
2	Boiler 1 Supply Water Temperature Setpoint	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:07 AM EST
3	Boiler 1 Status	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:07 AM EST
4	Heating Water Pump HWP-1 Enable	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:07 AM EST
5	Heating Water Pump HWP-1 Status	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/22/2023 10:08 AM EST

#	Question	Answer	Details
6	Heating Water Supply Temp	☐ Pass ☐ Fail ☒ N/A	
		Checked by: Theo Lagonikos on 02/22/2023 10:08 AM EST	
7	Boiler Entering Water Temperature	☐ Pass ☐ Fail ☒ N/A	
		Checked by: Theo Lagonikos on 02/22/2023 10:08 AM EST	
8	Heating Water Return Temp	☐ Pass ☐ Fail ☒ N/A	
		Checked by: Theo Lagonikos on 02/22/2023 10:08 AM EST	
9	Heating Water System Load	☐ Pass ☐ Fail ☒ N/A	
		Checked by: Theo Lagonikos on 02/22/2023 10:08 AM EST	

ONSITE FUNCTIONAL TESTING

Power Disconnect

- Verify that the equipment is provided with an easily accessible power disconnect. Verify that the disconnect is mounted directly to the equipment it serves or is located near the equipment it serves.

☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/22/2023 10:13 AM EST
- If the equipment is served by a disconnect that is mounted separately, verify that a permanent placard is affixed to the power disconnect serving this equipment and that the placard ID matches the placard affixed to the equipment.

☐ Pass ☐ Fail ☒ N/A

Checked by: Theo Lagonikos on 02/22/2023 10:13 AM EST
- With the disconnect in the ON position, verify that moving the disconnect to the OFF position removes all power from the equipment. Consult with appropriate contractors prior to energizing or de-energizing any equipment.

☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/22/2023 10:17 AM EST
- With the disconnect in the OFF position, verify that moving the disconnect to the ON position enables power to equipment and that the equipment energizes properly. Consult with appropriate contractors prior to energizing or de-energizing any equipment.

☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/22/2023 10:18 AM EST
- Perform this check for each piece of equipment associated with this system that is provided with a power disconnect.

☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/22/2023 10:51 AM EST

#	Question	Answer	Details
Alarms - Confirm the following alarms are programmed and display on the BAS when initiated			
1	Simulate Boiler 1 status not matching command operation and verify: Boiler 1 Status Alarm is generated in the BAS.	☐ Pass ☒ Fail ☐ N/A	No status currently for boiler. Need CT for boiler status
Checked by: Theo Lagonikos on 02/22/2023 10:54 AM EST			
2	Simulate a Boiler 1 heating water supply temperature more than 10°F above setpoint and verify: Boiler 1 Leaving Water Temperature High Alarm is generated in the BAS after 10 minutes.	☐ Pass ☒ Fail ☐ N/A	Currently does not alarm. Can be programmed based on boiler integral setting of 160.
Checked by: Theo Lagonikos on 02/22/2023 10:53 AM EST			
3	Simulate a Boiler 1 heating water return temperature less than 10°F below heating water supply temperature and verify: Boiler 1 Leaving Water Temperature Low Alarm is generated in the BAS after 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No alarm for HWR. Boiler does not modulate
Checked by: Theo Lagonikos on 02/22/2023 10:44 AM EST			
4	Simulate a Boiler 1 heating water return temperature more than 50°F below heating water supply temperature and verify: Boiler 1 Entering Water Temperature Low Alarm is generated in the BAS after 10 minutes.	☐ Pass ☒ Fail ☐ N/A	No alarm for HWR. Boiler does not modulate
Checked by: Theo Lagonikos on 02/22/2023 10:44 AM EST			
5	Simulate a primary boiler pump failure and verify: Boiler pump and the associated boiler disable. Boiler Pump Failure Alarm is generated in the BAS. Manual reset is required prior to pump staging to return to automatic operation.	☐ Pass ☒ Fail ☐ N/A	Pump and boiler disabled. No alarm currently programmed
Checked by: Theo Lagonikos on 02/22/2023 10:51 AM EST			
6	Simulate Heating Water Pump status not matching command operation and verify: Heating Water Pump Status Alarm is generated in the BAS.	☐ Pass ☒ Fail ☐ N/A	No alarms currently programmed. HWP does have status
Checked by: Theo Lagonikos on 02/22/2023 10:56 AM EST			
Off Mode			
1	Schedule the system into the Off mode and verify: 1. Heating water system is disabled. 2. Boiler is disabled. Boiler pump continues to run for 5 minutes. 3. Heating water pump is disabled.	☒ Pass ☐ Fail ☐ N/A	
Checked by: Theo Lagonikos on 02/22/2023 10:25 AM EST			
On Mode			

#	Question	Answer	Details
1	Simulate outside air temperature below 70° (adj.) and verify: 1. Heating water system is enabled into the On mode. 2. Boiler pump enables, after a 30 second delay (adj.) the boiler enables. 3. Heating water pump (HWP-1) is energized. 4. Integral boiler controller modulates boiler firing range and staging to maintain the heating water supply temperature setpoint.	☒ Pass ☐ Fail ☐ N/A	1 stage. Does not modulate

Checked by: Theo Lagonikos on 02/22/2023 10:22 AM EST

Heating Water Pump Control

1	While heating water system is enabled, verify: 1. Heating water pump is enabled. 2. Pump starts and runs continuously at constant speed.	☒ Pass ☐ Fail ☐ N/A	
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Checked by: Theo Lagonikos on 02/22/2023 10:21 AM EST

Return Equipment/System to Normal

1	Return all equipment/system components back to their normal and automatic operating conditions.	☒ Pass ☐ Fail ☐ N/A	
2	Ensure all BAS control points that were placed into manual override during testing are released back to their normal states/values.	☒ Pass ☐ Fail ☐ N/A	

Checked by: Theo Lagonikos on 02/22/2023 10:52 AM EST

Related Issues

#: 12	Asset: Hot Water System	Code: FPT-34-001	Status: Open	Priority: Low
Reported By: Theo Lagonikos on 02/22/2023				
Assigned to : Unassigned				
Description: Boiler status not available. Will need to add current transducer to get boiler status				
Response(s): None				

END TEST

FPT-60: (E) UH-1-H (GF UH)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) UH-1-H
Type	UNIT HEATER
Description	UNIT HEATER
Discipline	HVAC
Location(s)	HANGAR>PUMP RM 100B





Sequence Of Operation

UNIT HEATER (UH 1-7)

1. THE UNIT HEATER SHALL AUTOMATICALLY ENABLE/DISABLE AND HEAT VIA LOCAL THERMOSTAT TO MAINTAIN SPACE TEMPERATURE SETPOINT.

Diagrams









Spirit Airlines
IAH - Warehouse and Hanger

(E) UH-1-H (GF UH)
FPT-60

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Temperature Setpoint ☒ Pass ☐ Fail ☐ N/A 60 deg F

Checked by: Theo Lagonikos on 03/06/2023 10:42 AM EST

Sensors -
Confirm the installed location of each sensor is near the design location. Confirm if unit is displayed at BMS, if so, record current reading .

1 Space Temperature Sensor ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 03/06/2023 10:42 AM EST

Devices and Outputs -
Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off)

1 UH Command (On/Off) ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 03/06/2023 10:42 AM EST

Unit Heater - Gas Functional Performance Testing

Operation

1 Simulate the space temperature below the space temperature setpoint and verify:
1. Unit is enabled
2. Unit modulates to maintain space temperature setpoint.
3. Once space temperature is above space setpoint, unit is disabled. ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 03/06/2023 10:42 AM EST

END TEST

Spirit Airlines
IAH - Warehouse and Hanger

(E) UH-2-H (GF UH)
FPT-61

FPT-61: (E) UH-2-H (GF UH)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) UH-2-H
Type	UNIT HEATER
Description	UNIT HEATER
Discipline	HVAC
Location(s)	HANGAR>BATHROOM 100A



Sequence Of Operation

UNIT HEATER (UH 1-7)

1. THE UNIT HEATER SHALL AUTOMATICALLY ENABLE/DISABLE AND HEAT VIA LOCAL THERMOSTAT TO MAINTAIN SPACE TEMPERATURE SETPOINT.

Diagrams





Spirit Airlines
IAH - Warehouse and Hanger

(E) UH-2-H (GF UH)
FPT-61

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Temperature Setpoint ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 03/06/2023 10:48 AM EST

Sensors -
Confirm the installed location of each sensor is near the design location. Confirm if unit is displayed at BMS, if so, record current reading .

1 Space Temperature Sensor ☒ Pass ☐ Fail ☐ N/A Electric unit heater has integral thermostat

Checked by: Theo Lagonikos on 03/06/2023 10:48 AM EST

Devices and Outputs -
Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off)

1 UH Command (On/Off) ☐ Pass ☒ Fail ☐ N/A

Checked by: Theo Lagonikos on 03/06/2023 10:48 AM EST

Unit Heater - Gas Functional Performance Testing

Operation

1 Simulate the space temperature below the space temperature setpoint and verify:
1. Unit is enabled
2. Unit modulates to maintain space temperature setpoint.
3. Once space temperature is above space setpoint, unit is disabled. ☐ Pass ☒ Fail ☐ N/A

Checked by: Theo Lagonikos on 03/06/2023 10:48 AM EST

Spirit Airlines
IAH - Warehouse and Hanger

(E) UH-2-H (GF UH)
FPT-61

Related Issues

#: 30 Asset: (E) UH-2-H Code: FPT-61-001 Status: Open Priority: Low

Reported By: Theo Lagonikos on 03/06/2023

Location(s): HANGAR>BATHROOM 100A

Assigned to : Unassigned

Description:

Electric unit heater fan did not enable. Electric resistant heating element is operational.

Response(s):

None

END TEST

FPT-37: (E) UH-3-W (GF UH)

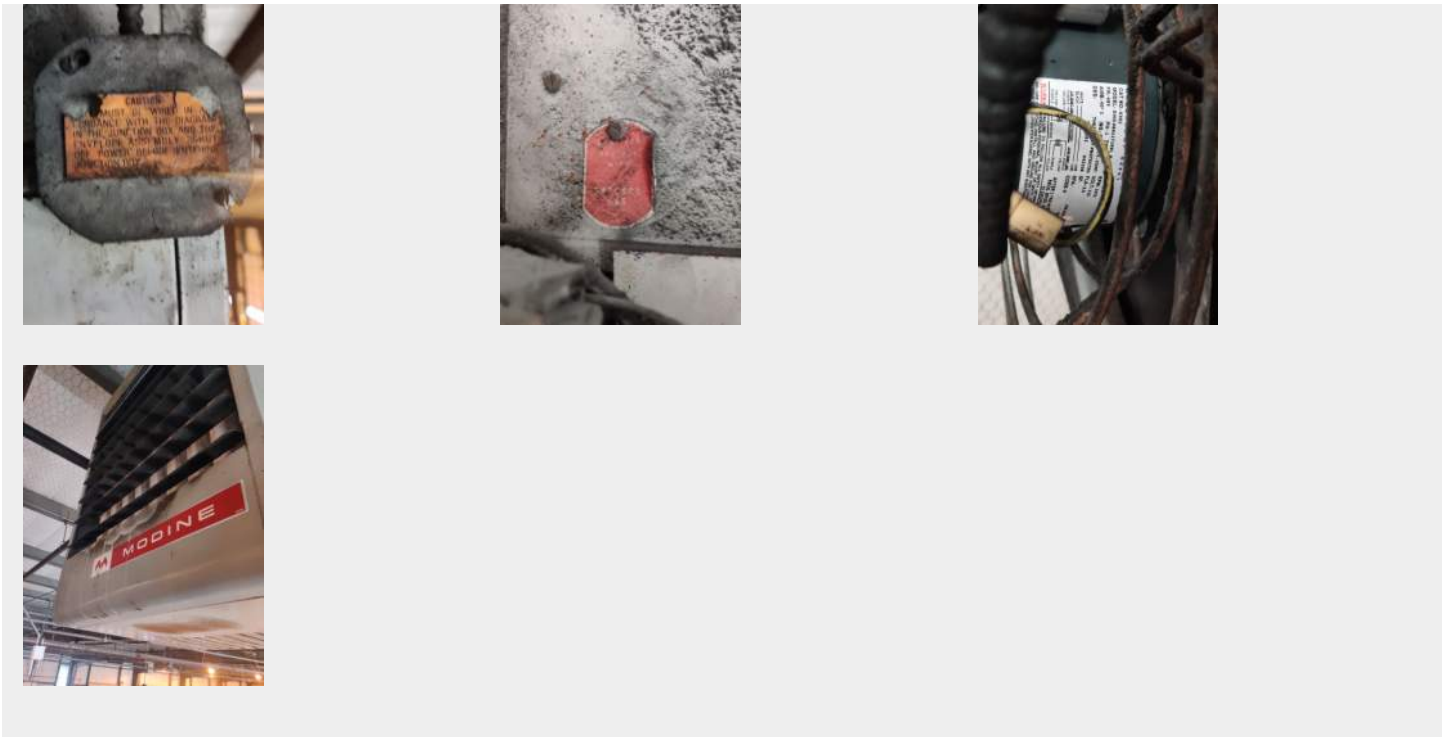
Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) UH-3-W
Type	UNIT HEATER
Description	UNIT HEATER
Discipline	HVAC
Location(s)	WAREHOUSE>TRUSS LEVEL





Sequence Of Operation

UNIT HEATER (UH 1-7)

1. THE UNIT HEATER SHALL AUTOMATICALLY ENABLE/DISABLE AND HEAT VIA LOCAL THERMOSTAT TO MAINTAIN SPACE TEMPERATURE SETPOINT.

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

Setpoints -

Record actual setpoint before starting the test (Submitted value in parentheses)

#	Question	Answer	Details
1	Space Temperature Setpoint	☒ Pass ☐ Fail ☐ N/A	Low setting or off. 60 deg approx (or off)



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Checked by: Theo Lagonikos on 02/21/2023 09:48 AM EST

Sensors –

Confirm the installed location of each sensor is near the design location. Confirm if unit is displayed at BMS, if so, record current reading .

1	Space Temperature Sensor	☒ Pass ☐ Fail ☐ N/A	
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Checked by: Theo Lagonikos on 02/21/2023 09:48 AM EST

Devices and Outputs –

Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off)

1	UH Command (On/Off)	☒ Pass ☐ Fail ☐ N/A	
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Checked by: Theo Lagonikos on 02/21/2023 09:48 AM EST

Unit Heater - Gas Functional Performance Testing

Operation

#	Question	Answer	Details
1	Simulate the space temperature below the space temperature setpoint and verify: 1. Unit is enabled 2. Unit modulates to maintain space temperature setpoint. 3. Once space temperature is above space setpoint, unit is disabled.	☒ Pass ☐ Fail ☐ N/A	



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Checked by: Theo Lagonikos on 02/21/2023 09:48 AM EST

END TEST

Spirit Airlines
IAH - Warehouse and Hanger

(E) UH-4-W (GF UH)
FPT-38

FPT-38: (E) UH-4-W (GF UH)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) UH-4-W
Type	UNIT HEATER
Description	UNIT HEATER
Discipline	HVAC
Location(s)	WAREHOUSE>TRUSS LEVEL



Sequence Of Operation

UNIT HEATER (UH 1-7)

1. THE UNIT HEATER SHALL AUTOMATICALLY ENABLE/DISABLE AND HEAT VIA LOCAL THERMOSTAT TO MAINTAIN SPACE TEMPERATURE SETPOINT.

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

#	Question	Answer	Details
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Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

END TEST

1 Space Temperature Setpoint ☒ Pass ☐ Fail ☐ N/A Low setting or off. 60 deg approx (or off). Shares tstat with UH-3

Checked by: Theo Lagonikos on 02/21/2023 09:49 AM EST

Sensors -
Confirm the installed location of each sensor is near the design location. Confirm if unit is displayed at BMS, if so, record current reading.

1 Space Temperature Sensor ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 09:49 AM EST

Devices and Outputs -
Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off)

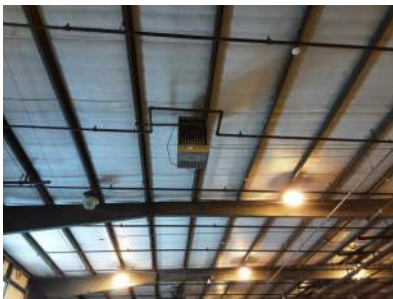
1 UH Command (On/Off) ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 09:49 AM EST

Unit Heater - Gas Functional Performance Testing

Operation

1 Simulate the space temperature below the space temperature setpoint and verify:
1. Unit is enabled
2. Unit modulates to maintain space temperature setpoint.
3. Once space temperature is above space setpoint, unit is disabled. ☒ Pass ☐ Fail ☐ N/A



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Checked by: Theo Lagonikos on 02/21/2023 09:49 AM EST

FPT-39: (E) UH-5-W (GF UH)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) UH-5-W
Type	UNIT HEATER
Description	UNIT HEATER
Discipline	HVAC
Location(s)	WAREHOUSE>TRUSS LEVEL





Sequence Of Operation

UNIT HEATER (UH 1-7)

1. THE UNIT HEATER SHALL AUTOMATICALLY ENABLE/DISABLE AND HEAT VIA LOCAL THERMOSTAT TO MAINTAIN SPACE TEMPERATURE SETPOINT.

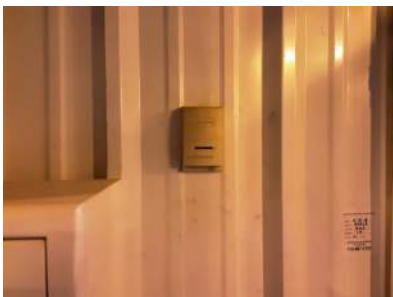
Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

Setpoints -

Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Temperature Setpoint ☒ Pass ☐ Fail ☐ N/A Low setting. 60 deg or off



1676991735872.jpg

Checked by: Theo Lagonikos on 02/21/2023 10:01 AM EST

Sensors -

Confirm the installed location of each sensor is near the design location. Confirm if unit is displayed at BMS, if so, record current reading .

#	Question	Answer	Details
1	Space Temperature Sensor	☒ Pass ☐ Fail ☐ N/A	

Checked by: Theo Lagonikos on 02/21/2023 10:01 AM EST

Devices and Outputs –
Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off)

1	UH Command (On/Off)	☒ Pass ☐ Fail ☐ N/A	
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Checked by: Theo Lagonikos on 02/21/2023 10:01 AM EST

Unit Heater - Gas Functional Performance Testing

Operation

1	Simulate the space temperature below the space temperature setpoint and verify: 1. Unit is enabled 2. Unit modulates to maintain space temperature setpoint. 3. Once space temperature is above space setpoint, unit is disabled.	☒ Pass ☐ Fail ☐ N/A	
---	--	---	--



1676991762599.jpg

Checked by: Theo Lagonikos on 02/21/2023 10:02 AM EST

END TEST

Spirit Airlines
IAH - Warehouse and Hanger

(E) UH-6-W (GF UH)
FPT-40

FPT-40: (E) UH-6-W (GF UH)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) UH-6-W
Type	UNIT HEATER
Description	UNIT HEATER
Discipline	HVAC
Location(s)	WAREHOUSE>TRUSS LEVEL



Sequence Of Operation

UNIT HEATER (UH 1-7)

1. THE UNIT HEATER SHALL AUTOMATICALLY ENABLE/DISABLE AND HEAT VIA LOCAL THERMOSTAT TO MAINTAIN SPACE TEMPERATURE SETPOINT.

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

#	Question	Answer	Details
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Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Temperature Setpoint ☒ Pass ☐ Fail ☐ N/A Shares tstat with uh-5

Checked by: Theo Lagonikos on 02/21/2023 10:03 AM EST

Sensors -
Confirm the installed location of each sensor is near the design location. Confirm if unit is displayed at BMS, if so, record current reading .

1 Space Temperature Sensor ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 10:03 AM EST

Devices and Outputs -
Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off)

1 UH Command (On/Off) ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 10:04 AM EST

Unit Heater - Gas Functional Performance Testing

Operation

1 Simulate the space temperature below the space temperature setpoint and verify:
1. Unit is enabled
2. Unit modulates to maintain space temperature setpoint.
3. Once space temperature is above space setpoint, unit is disabled. ☐ Pass ☒ Fail ☐ N/A Heater did not enable. Fan is operational



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Checked by: Theo Lagonikos on 02/21/2023 10:05 AM EST

Spirit Airlines
IAH - Warehouse and Hanger

(E) UH-6-W (GF UH)
FPT-40

Related Issues

#: 5 Asset: (E) UH-6-W Code: FPT-40-001 Status: Open Priority: Low

Reported By: Theo Lagonikos on 02/21/2023

Assigned to : Unassigned

Description:
Heater did not enable on UH-6

Response(s):
None

END TEST

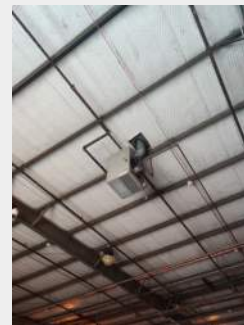
FPT-41: (E) UH-1-W (GF UH)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) UH-1-W
Type	UNIT HEATER
Description	UNIT HEATER
Discipline	HVAC
Location(s)	WAREHOUSE>TRUSS LEVEL



Sequence Of Operation

UNIT HEATER (UH 1-7)

1. THE UNIT HEATER SHALL AUTOMATICALLY ENABLE/DISABLE AND HEAT VIA LOCAL THERMOSTAT TO MAINTAIN SPACE TEMPERATURE SETPOINT.

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

Spirit Airlines
IAH - Warehouse and Hanger

(E) UH-1-W (GF UH)
FPT-41

#	Question	Answer	Details
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Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Temperature Setpoint ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/22/2023 10:59 AM EST

Sensors -
Confirm the installed location of each sensor is near the design location. Confirm if unit is displayed at BMS, if so, record current reading .

1 Space Temperature Sensor ☒ Pass ☐ Fail ☐ N/A



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Checked by: Theo Lagonikos on 02/21/2023 09:31 AM EST

Devices and Outputs -
Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off)

1 UH Command (On/Off) ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 09:33 AM EST

Unit Heater - Gas Functional Performance Testing

Operation

#	Question	Answer	Details
1	Simulate the space temperature below the space temperature setpoint and verify: 1. Unit is enabled 2. Unit modulates to maintain space temperature setpoint. 3. Once space temperature is above space setpoint, unit is disabled.	☒ Pass ☐ Fail ☐ N/A	200 deg F SAT



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Checked by: Theo Lagonikos on 02/21/2023 09:35 AM EST

END TEST

Spirit Airlines
IAH - Warehouse and Hanger

(E) UH-2-W (GF UH)
FPT-42

FPT-42: (E) UH-2-W (GF UH)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) UH-2-W
Type	UNIT HEATER
Description	UNIT HEATER
Discipline	HVAC
Location(s)	WAREHOUSE>TRUSS LEVEL



Sequence Of Operation

UNIT HEATER (UH 1-7)

1. THE UNIT HEATER SHALL AUTOMATICALLY ENABLE/DISABLE AND HEAT VIA LOCAL THERMOSTAT TO MAINTAIN SPACE TEMPERATURE SETPOINT.

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

#	Question	Answer	Details
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Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

END TEST

1 Space Temperature Setpoint ☒ Pass ☐ Fail ☐ N/A Low setting or off. Shares tstat with UH-1

Checked by: Theo Lagonikos on 02/21/2023 09:50 AM EST

Sensors -
Confirm the installed location of each sensor is near the design location. Confirm if unit is displayed at BMS, if so, record current reading .

1 Space Temperature Sensor ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 09:40 AM EST

Devices and Outputs -
Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off)

1 UH Command (On/Off) ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 09:40 AM EST

Unit Heater - Gas Functional Performance Testing

Operation

1 Simulate the space temperature below the space temperature setpoint and verify:
1. Unit is enabled
2. Unit modulates to maintain space temperature setpoint.
3. Once space temperature is above space setpoint, unit is disabled. ☒ Pass ☐ Fail ☐ N/A



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Checked by: Theo Lagonikos on 02/21/2023 09:40 AM EST

FPT-43: (E) UH-7-W (GF UH)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) UH-7-W
Type	UNIT HEATER
Description	UNIT HEATER
Discipline	HVAC
Location(s)	WAREHOUSE>TRUSS LEVEL



Sequence Of Operation

UNIT HEATER (UH 1-7)

1. THE UNIT HEATER SHALL BE AUTOMATICALLY ENABLED/DISABLED BY THE DDC SYSTEM CONTROLLER WHENEVER THE HOA SWITCH IS IN THE AUTOMATIC POSITION AND MANUALLY STARTED AND STOPPED BY THE HAND POSITION. WHEN ENABLED, THE UNIT HEATER SHALL OPERATE UNDER ITS OWN CONTROLS TO MAINTAIN SPACE TEMPERATURE OF 40° F (ADJ.)

Questionnaire

#	Question	Answer	Details
Point-to-Point Checkout			

Setpoints -
Record actual setpoint before starting the test (Submitted value in parentheses)

1 Space Temperature Setpoint ☒ Pass ☐ Fail ☐ N/A Low setting. 60 deg. Auto. Heat.



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Checked by: Theo Lagonikos on 02/21/2023 09:58 AM EST

Sensors -
Confirm the installed location of each sensor is near the design location. Confirm if unit is displayed at BMS, if so, record current reading.

1 Space Temperature Sensor ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 09:59 AM EST

Devices and Outputs -
Review physical location and confirm operation of each device. Manually command the device through the range of its operation (On/Off)

1 UH Command (On/Off) ☒ Pass ☐ Fail ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 09:59 AM EST

Unit Heater - Gas Functional Performance Testing

#	Question	Answer	Details
Operation			
1	Simulate the space temperature below the space temperature setpoint and verify: 1. Unit is enabled 2. Unit modulates to maintain space temperature setpoint. 3. Once space temperature is above space setpoint, unit is disabled.	☒ Pass ☐ Fail ☐ N/A	



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Checked by: Theo Lagonikos on 02/21/2023 09:59 AM EST

END TEST

FPT-44: (E) EF-1-W (Toilet Exhaust)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) EF-1-W
Type	EXHAUST FAN
Description	EXHAUST FAN
Discipline	HVAC
Location(s)	WAREHOUSE>REST RM MENS



System Description



Spirit Airlines
IAH - Warehouse and Hanger

(E) EF-1-W (Toilet Exhaust)
FPT-44

Toilet exhaust fan interlocked with AHU-1-W.

Questionnaire

#	Question	Answer	Details
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Sequence of Operation

1 Verify if EF-1-W is interlocked with AHU-1-W ☒ Pass ☐ Fail ☐ N/A EF-1 is interlocked to AHU-1
Checked by: Theo Lagonikos on 02/21/2023 11:01 AM EST

Enable

1 Exhaust fan shall turn on ☒ Yes ☐ No ☐ N/A
Checked by: Theo Lagonikos on 02/21/2023 10:57 AM EST

Disable AHU-1-W

1 Exhaust fan shall turn off ☒ Yes ☐ No ☐ N/A
Checked by: Theo Lagonikos on 02/21/2023 11:00 AM EST

END TEST

FPT-54: (E) CRAC-1 (DX CRAC)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) CRAC-1
Type	COMPUTER RM AIR CONDITIONER
Description	COMPUTER RM AIR CONDITIONER
Discipline	HVAC
Location(s)	HANGAR>IT RM 231



System Description

Computer Room Air Conditioning (CRAC) - Each CRAC unit is controlled independently via the onboard controller.

Sequence Of Operation

See approved control submittal

Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Unoccupied Cooling Setpoint					
Occupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					
Unoccupied Humidity Setpoint					
Occupied Humidity Setpoint					

Sensor Calibration

Sensor & Location	Location OK?	First Gauge or BAS Value	Instrument Measured Value	Final Gauge or BAS Value	Pass
SPACE/RETURN TEMPERATURE					
SPACE/RETURN HUMIDITY					
SUPPLY AIR TEMPERATURE					

Device Calibration

Device or Actuator & Location	Type	Procedure/ State	First BAS Value	Device State/ Display Readout	Final BAS Value	Pass
FAN CONTROL	Two Position(On-Off)	On				
		Off				

Device or Actuator & Location	Type	Procedure/ State	First BAS Value	Device State/ Display Readout	Final BAS Value	Pass
COOLING CONTROL	Modulating	Full Open				
		% Open				
		Closed				
HEATING CONTROL	Modulating	Full Open				
		% Open				
		Closed				
HUMIDIFIER CONTROL	Modulating	Full Open				
		% Open				
		Closed				

Questionnaire

#	Question	Answer	Details
Safeties			
1	Supply Air Smoke Detector Alarm	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 03/20/2023 02:21 PM EDT
2	Building Fire Alarm	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 03/20/2023 02:21 PM EDT
3	High condensate	☐ Pass ☒ Fail ☐ N/A	Could not test Checked by: Theo Lagonikos on 03/20/2023 02:23 PM EDT

Alarms

IF MONITORED BY BMS:
Simulate alarms before testing the sequence of operation. The unit should remain "On" and an alarm should be present on the BMS.



Spirit Airlines
IAH - Warehouse and Hanger

(E) CRAC-1 (DX CRAC)
FPT-54

#	Question	Answer	Details
1	Supply fans command and status not matching	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/21/2023 04:27 PM EST
2	Supply air temperature	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/21/2023 04:28 PM EST
3	Return/Space air temperature	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/21/2023 04:28 PM EST

Operation Modes

Disable

1	Disable the unit via local controller (or from BMS if integrated). Verify: -The fan is off -The associated condensing unit is off.	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/21/2023 03:57 PM EST
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Enable

1	Enable the unit. Verify: -The fan is on -The associated condensing unit is on.	☐ Pass ☒ Fail ☐ N/A	Could not test. Unit not fully operational. Checked by: Theo Lagonikos on 03/20/2023 02:21 PM EDT
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Control

Cooling

1	Record current space temperature setpoint	☒ Pass ☐ Fail ☐ N/A	65F Checked by: Theo Lagonikos on 02/21/2023 04:27 PM EST
2	Record current relative humidity setpoint	☒ Pass ☐ Fail ☐ N/A	55% Checked by: Theo Lagonikos on 02/21/2023 04:27 PM EST
3	Enable the unit and lower the space temperature setpoint below the the occupied cooling setpoint. Verify: -The fan is on -The cooling control modulates to maintain the space temperate setpoint.	☐ Pass ☒ Fail ☐ N/A	Could not test. Unit not fully operational. Checked by: Theo Lagonikos on 03/20/2023 02:21 PM EDT

#	Question	Answer	Details
4	Record discharge air temperature with temperature gun.	☐ Pass ☒ Fail ☐ N/A	
			Checked by: Theo Lagonikos on 03/20/2023 02:21 PM EDT
5	Increase temperature setpoint above space temperature. Verify: -The fan is off	☐ Pass ☒ Fail ☐ N/A	
			Checked by: Theo Lagonikos on 03/20/2023 02:21 PM EDT

Humidity Control

1	Record discharge air humidity during normal operation from the local control panel or hand-held hygrometer.	☒ Pass ☐ Fail ☐ N/A	Space at 58%
			Checked by: Theo Lagonikos on 02/21/2023 04:46 PM EST
2	Increase humidity setpoint above current setpoint. Verify humidifier is operational by confirming discharge air humidity on the local control panel or with a hand-held hygrometer is higher than normal operation.	☐ Pass ☒ Fail ☐ N/A	Could not test. Unit not fully operational.
			Checked by: Theo Lagonikos on 03/20/2023 02:21 PM EDT

Trends

1	Trends show the equipment is maintaining the space temperature setpoint.	☐ Pass ☐ Fail ☒ N/A	
			Checked by: Theo Lagonikos on 02/21/2023 04:47 PM EST
2	Trends show the equipment is maintaining the space humidity setpoint.	☐ Pass ☐ Fail ☒ N/A	
			Checked by: Theo Lagonikos on 02/21/2023 04:47 PM EST

End of Test

Release all overrides and modification used during testing.

1	The returns to normal operation.	☐ Pass ☐ Fail ☒ N/A	Unit could not be fully tested.
			Checked by: Zach Yohon on 03/20/2023 12:06 PM EDT

Spirit Airlines
IAH - Warehouse and Hanger

(E) CRAC-1 (DX CRAC)
FPT-54

Related Issues

#: 10 Asset: (E) CRAC-1 Code: FPT-54-001 Status: Open Priority: Low

Reported By: Theo Lagonikos on 02/21/2023

Assigned to : Unassigned

Description:

Unit compressors are not enabling. Potential electrical issues. Display board not functioning.

Response(s):

None

END TEST

FPT-55: (E) CRAC-2 (DX CRAC)

Test Type: Functional Performance Testing

Status: Failed

Asset Summary

Unit #	(E) CRAC-2
Type	COMPUTER RM AIR CONDITIONER
Description	COMPUTER RM AIR CONDITIONER
Discipline	HVAC
Location(s)	HANGAR>IT RM 231



System Description

Computer Room Air Conditioning (CRAC) - Each CRAC unit is controlled independently via the onboard controller.

Sequence Of Operation

See approved control submittal

Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Unoccupied Cooling Setpoint					
Occupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					
Unoccupied Humidity Setpoint					
Occupied Humidity Setpoint					

Sensor Calibration

Sensor & Location	Location OK?	First Gauge or BAS Value	Instrument Measured Value	Final Gauge or BAS Value	Pass
SPACE/RETURN TEMPERATURE					
SPACE/RETURN HUMIDITY					
SUPPLY AIR TEMPERATURE					

Device Calibration

Device or Actuator & Location	Type	Procedure/ State	First BAS Value	Device State/ Display Readout	Final BAS Value	Pass
FAN CONTROL	Two Position(On-Off)	On				
		Off				

Device or Actuator & Location	Type	Procedure/ State	First BAS Value	Device State/ Display Readout	Final BAS Value	Pass
COOLING CONTROL	Modulating	Full Open				
		% Open				
		Closed				
HEATING CONTROL	Modulating	Full Open				
		% Open				
		Closed				
HUMIDIFIER CONTROL	Modulating	Full Open				
		% Open				
		Closed				

Questionnaire

#	Question	Answer	Details
Safeties			
1	Supply Air Smoke Detector Alarm	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 03/20/2023 02:22 PM EDT
2	Building Fire Alarm	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 03/20/2023 02:22 PM EDT
3	High condensate	☐ Pass ☒ Fail ☐ N/A	Could not test. Unit not fully operational. Checked by: Theo Lagonikos on 03/20/2023 02:22 PM EDT

Alarms

IF MONITORED BY BMS:
Simulate alarms before testing the sequence of operation. The unit should remain "On" and an alarm should be present on the BMS.



Spirit Airlines
IAH - Warehouse and Hanger

(E) CRAC-2 (DX CRAC)
FPT-55

#	Question	Answer	Details
1	Supply fans command and status not matching	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/21/2023 05:00 PM EST
2	Supply air temperature	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/21/2023 05:00 PM EST
3	Return/Space air temperature	☐ Pass ☐ Fail ☒ N/A	Checked by: Theo Lagonikos on 02/21/2023 05:00 PM EST

Operation Modes

Disable

1	Disable the unit via local controller (or from BMS if integrated). Verify: -The fan is off -The associated condensing unit is off.	☒ Pass ☐ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/21/2023 05:02 PM EST
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Enable

1	Enable the unit. Verify: -The fan is on -The associated condensing unit is on.	☐ Pass ☒ Fail ☐ N/A	Could not test. Unit not fully operational. Checked by: Theo Lagonikos on 03/20/2023 02:22 PM EDT
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Control

Cooling

1	Record current space temperature setpoint	☒ Pass ☐ Fail ☐ N/A	72F Checked by: Theo Lagonikos on 02/21/2023 05:01 PM EST
2	Record current relative humidity setpoint	☒ Pass ☐ Fail ☐ N/A	57% Checked by: Theo Lagonikos on 02/21/2023 05:01 PM EST
3	Enable the unit and lower the space temperature setpoint below the the occupied cooling setpoint. Verify: -The fan is on -The cooling control modulates to maintain the space temperate setpoint.	☐ Pass ☒ Fail ☐ N/A	Checked by: Theo Lagonikos on 02/21/2023 05:23 PM EST

#	Question	Answer	Details
4	Record discharge air temperature with temperature gun.	☐ Pass ☒ Fail ☐ N/A	Unit not working correctly, could not read discharge air temp. Checked by: Theo Lagonikos on 03/20/2023 02:22 PM EDT
5	Increase temperature setpoint above space temperature. Verify: -The fan is off	☐ Pass ☒ Fail ☐ N/A	Could not test. Unit not fully operational. Checked by: Theo Lagonikos on 03/20/2023 02:22 PM EDT

Humidity Control

1	Record discharge air humidity during normal operation from the local control panel or hand-held hygrometer.	☒ Pass ☐ Fail ☐ N/A	57% Checked by: Theo Lagonikos on 02/21/2023 05:24 PM EST
2	Increase humidity setpoint above current setpoint. Verify humidifier is operational by confirming discharge air humidity on the local control panel or with a hand-held hygrometer is higher than normal operation.	☐ Pass ☒ Fail ☐ N/A	Could not test. Unit not fully operational. Checked by: Theo Lagonikos on 03/20/2023 02:22 PM EDT

Trends

1	Trends show the equipment is maintaining the space temperature setpoint.	☐ Pass ☐ Fail ☒ N/A	 Checked by: Theo Lagonikos on 02/21/2023 05:01 PM EST
2	Trends show the equipment is maintaining the space humidity setpoint.	☐ Pass ☐ Fail ☒ N/A	 Checked by: Theo Lagonikos on 02/21/2023 05:01 PM EST

End of Test

Release all overrides and modification used during testing.

1	The returns to normal operation.	☐ Pass ☒ Fail ☐ N/A	Unit could not be fully tested. Checked by: Theo Lagonikos on 03/20/2023 02:22 PM EDT
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Spirit Airlines
IAH - Warehouse and Hanger

(E) CRAC-2 (DX CRAC)
FPT-55

Related Issues

#: 11 Asset: (E) CRAC-2 Code: FPT-55-001 Status: Open Priority: Low

Reported By: Theo Lagonikos on 02/21/2023

Assigned to : Unassigned

Description:

Unit compressors are not enabling. Potential electrical issues. Display board not functioning.

Response(s):

None

END TEST

FPT-56: (E) AHU-2 (RCx - AHU-2)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) AHU-2
Type	AIR HANDLING UNIT
Description	AIR HANDLING UNIT
Discipline	HVAC
Location(s)	HANGAR>ADMIN ROOF



System Description



Spirit Airlines
IAH - Warehouse and Hanger

(E) AHU-2 (RCx - AHU-2)
FPT-56

DX Cooling only AHU serving 2nd Floor IT Room.

Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Unoccupied Cooling Setpoint					
Occupied Cooling Setpoint					
Occupied Heating Setpoint					
Unoccupied Heating Setpoint					

Sensor Calibration

Sensor & Location	Location OK?	First Gauge or BAS Value	Instrument Measured Value	Final Gauge or BAS Value	Pass
SPACE TEMPERATURE					
FAN STATUS					

Device Calibration

Device or Actuator & Location		Type	Procedure/ State	First BAS Value	Device State/ Display Readout	Final BAS Value	Pass
OUTDOOR CONTROL	DAMPER	Two Position(On-Off)	On				
			Off				

Device or Actuator & Location	Type	Procedure/ State	First BAS Value	Device State/ Display Readout	Final BAS Value	Pass
HEATING CONTROL	Modulating	Full Open				
		% Open				
		Closed				
COOLING CONTROL	Modulating	Full Open				
		% Open				
		Closed				
FAN MOTOR COMMAND	Two Position(On-Off)	On				
		Off				
FAN MOTOR CONTROL	ECM	VDC				
		VDC				
		VDC				

Questionnaire

#	Question	Answer	Details
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Safeties

None

Control

Cooling

Simulate the unit in cooling mode by adjusting the space temperature setpoint below the space temperature.



Spirit Airlines
IAH - Warehouse and Hanger

(E) AHU-2 (RCx - AHU-2)
FPT-56

#	Question	Answer	Details
1	The AHU is enabled and the fan is energized, the heating valve is closed, the cooling valve modulates to maintain the space temperature setpoint	☒ Pass ☐ Fail ☐ N/A	

Checked by: Theo Lagonikos on 02/21/2023 05:20 PM EST

End of Test

Release all overrides and modification used during testing.

1	The returns to normal operation.	☒ Pass ☐ Fail ☐ N/A	
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Checked by: Theo Lagonikos on 02/21/2023 05:20 PM EST

END TEST

FPT-57: (E) HV-6 (Dx FCU)

Test Type: Functional Performance Testing

Status: Completed

Asset Summary

Unit #	(E) HV-6
Type	HEATING VENTILLATOR UNIT
Description	HEATING VENTILLATOR UNIT
Discipline	HVAC
Location(s)	HANGAR>STORAGE 235



Operational Assumptions

Operating Conditions:

1. BAS to control adjustable temperature set point.
2. Set thermostat at 5F above set temperature from split system setting.

Alarms and Safeties:

1. Generate an alarm when space temperature increase above 10F.

Set Points

Parameter	Description / Trend Information	Setpoint		Alarm	
		Design	Actual	Design	Actual
Zone temperature set point	75°F				

Questionnaire

#	Question	Answer	Details
Sequence of Operations			

- 1 Simulate the space temperature rising 5°F above the split system space temperature setpoint and verify:
 1. Split system energizes and controls to zone temperature setpoint
 2. The Outdoor Unit is indexed on
 3. Discharge air is cooling the space (verify with handheld infrared thermometer)
- ☒ Pass
 ☐ Fail
 ☐ N/A



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Checked by: Theo Lagonikos on 02/21/2023 03:42 PM EST

- 2 Simulate the space temperature falling to the split system space temperature setpoint and verify:
 1. Split system de-energizes
- ☒ Pass
 ☐ Fail
 ☐ N/A

Checked by: Theo Lagonikos on 02/21/2023 03:42 PM EST

Condensate Pump

Spirit Airlines
IAH - Warehouse and Hanger

(E) HV-6 (Dx FCU)
FPT-57

#	Question	Answer	Details
1	Verify operation of the condensate pump	☐ Pass ☐ Fail ☒ N/A	Gravity drain to outdoors



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Checked by: Theo Lagonikos on 02/21/2023 03:26 PM EST

Final Verification

1	Ensure all changed parameters, overrides are returned to normal.	☒ Pass ☐ Fail ☐ N/A	
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Checked by: Theo Lagonikos on 02/21/2023 03:42 PM EST

END TEST

